Conditionals: first steps

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1. Background

- **Sentences** versus **propositions**
  - Propositions: things that can be true or false; things we mean, assert, deny, believe, are more or less confident of…
  - For the proposition that snow is white to be true is for snow to be white
- **Context sensitivity**. A context-sensitive sentence is one that can be used, literally, to express multiple propositions.
- **Connectives**. In formal languages, we find expressions called *connectives*, each of which combines with a certain specified number of sentences to form a new sentence.
  - Familiar unary connectives: $\neg$, $\Diamond$, $\Box$
  - Familiar binary connectives: $\wedge$, $\vee$, $\supset$

  The meaning of an *n*-ary connective can be specified as a *function* from *n*-tuples of propositions to propositions. A context-insensitive connective is associated with a single such function. A context-sensitive one can be associated with different functions on different occasions.
- A **truth-functional** connective is one that expresses a function *f* from *n*-tuples of propositions to propositions with the following property: whenever P_1 and Q_1 are either both true or both false, and … and P_n and Q_n are either both true or both false, then $f(P_1,\ldots,P_n)$ and $f(Q_1,\ldots,Q_n)$ are either both true or both false. Not all connectives are truth functional: for example, even when ϕ and ψ are both true, $\Box\phi$ may be true and $\Box\psi$ false.

2. Conditionals

Our central focus: declarative English sentences of the following forms, and their meanings:

If ϕ, ψ If ϕ, then ψ ψ if ϕ

In each case we have an **if-clause**, ‘if ϕ’, and a **main clause**, ‘ψ’ (or ‘then ψ’)

(Some authors use ‘conditional’ for the sentences; others, for the propositions they express.)

However it is worth paying some attention to the various ways in which similar meanings can be expressed without 'if':

(1) a. Unless there has been an accident she is nearly home

b. Had we not won, we would have been in trouble

c. Whatever Bill says, Mary will quit her job

d. One more such victory and we are lost

e. My kids would love a pet dog

We should also not forget about conditional questions and imperatives:

(2) a. Where would you live if you were a millionaire?

b. If the butler did it, what weapon did he use?

c. If you're so smart, why aren't you rich?

d. Attack immediately if you spot the enemy.

e. If you're so smart, prove it yourself.

3. 'If' as a binary connective

There is a long tradition of treating 'if' as an expression of the same kind as 'and' or the '$\wedge$' of formal logic. On the obvious way of implementing this idea, we would expect the proposition expressed by 'If ϕ, ψ' in a given context *c* to be the result of applying the function expressed by 'If' in *c* to the ordered pair of the proposiiton expressed by ϕ in *c* and the proposiiton expressed by ψ in *c*: $[\![\text{If } \phi, \psi]\!]^c = [\![\text{If}]\!]^c([\![\phi]\!]^c, [\![\psi]\!]^c)$.

This idea faces several problems, some deeper than others.

Problem 0: Sometimes, including a context-sensitive sentence in a conditional makes it natural to interpret it differently from how it would have been interpreted if it had been uttered all by itself:

(3) If the White House basketball game took place yesterday, no tall people were at the meeting.

Response: remember that "contexts" here are theoretical entities. Distinguish *what sentence ϕ expresses in the context of utterance u* from *what would have been expressed by an utterance of ϕ if one had taken place instead of u*.

Problem 1: 'If ϕ, ψ', the 'if' clause 'if ϕ' is itself a meaningful constituent—an adverbial clause. (Compare: in 'I left because it was late', 'because it was late' is a meaningful constituent, whereas'I left because' is just a string of words.)

We can get around this problem by taking 'if ϕ' to have the meaning of a unary operator—a function from propositions to propositions—and 'if' to have the meaning of a function from propositions to such functions.

Problem 2: in

(4) If I resigned tomorrow, I would be hired by Google next week

it's quite unclear how to answer the question what propositions are expressed in the context by the constituent sentences 'I resigned tomorrow' and 'I would be hired by Google next week'. The former seems like nonsense; the latter is not nonsense, but it's got a kind of context-sensitivity that doesn't show up in the original sentence. What we'd like to say is that the proposition expressed by (***), as uttered by me now, is determined by applying a certain function to the proposition that I *will* resign tomorrow and the proposition that I *will* be hired by Google next week.

Response: all right. What we are really interested in here is a certain way of building one sentence out of two others, which isn't just a matter of combining them with the word 'if', but also sometimes involves making changes of tense and inserting 'would'. Our task is to explain how the meaning of the output of this operation depends on the meanings of its inputs.

Problem 3: What are the two propositions that would be inputs to the relevant function in a sentence like

(5) If a programmer lives with non-programmers, the programmer is frustrated

Certainly the main clause does not contribute the proposition that the one and only programmer (in the world, or in some limited domain) is frustrated, or that some particular person is frustrated, or anything like that! The question seems misconceived!

This issue is actually quite pervasive, especially with 'does-will' conditionals.

(5a) If Jack sees Jill next week, he will wave

What proposition would play the role of the consequent here? Not just the proposition that Jack will wave sometime or other, or the proposition that Jack will wave next week, or even the proposition that Jack will wave at Jill next week…

Let's come back to cases like these many weeks from now (when we look at Kratzer's theory). For now we will try to focus on cases where it's clear that we are dealing with two specific propositions.

Two kinds of conditionals, and two analyses

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1. Indicative versus counterfactual conditionals

A famous contrast (Adams 1970):

(1) If Oswald didn't kill Kennedy, someone else did

(2) If Oswald hadn't killed Kennedy, someone else would have

Insofar as we are buying into the two-proposition model, it's clear that these involve the same two propositions: the proposition that Oswald didn't kill Kennedy and the proposition that someone other than Oswald killed Kennedy. But they mean very different things.

(1) is called an *indicative* conditional; (2) a *counterfactual* or *subjunctive* conditional. We'll use the labels 'counterfactual' and 'subjunctive' interchangeably.

- A problematic suggestion: what's semantically distinctive about (2) is that it entails, or suggests, that it is *false* that Oswald killed Kennedy. Evidence against this (Anderson 1951):

(3) If Jones had taken arsenic, he would have shown just exactly those symptoms which he does in fact show.

  Thus 'counterfactual' is something of a misnomer.

- But so is 'subjunctive': (2) does not use the subjunctive mood (which is pretty vestigial in English). Its grammatical hallmarks are, rather: (a) the presence of 'would' in the main clause, and (b) an additional level of past tense in the 'if' clause. Sabrina Iatridou (2000) calls (b) "fake past tense", and finds analogues in many other languages — see section 6.2 of the von Fintel handbook article.
- There is a debate about whether to classify (4) along with (1) or along with (2):

(4) If Oswald doesn't kill Kennedy, someone else will.

  See Bennett §6. Given this controversy, let's try whenever possible to avoid relying on such 'does-will' conditionals as examples.

  Incidentally: the following contrast may help shed light on this dispute:

(5) a. If you were given a million dollars tomorrow, you would be richer than you will ever actually be.

  b. ? If you are given a million dollars tomorrow, you will be richer than you will ever actually be.

- It’s worth considering what happens when we use different combnations of tenses and ‘would’ marking:

(6) a. If Oswald hadn’t killed Kennedy, someone else had

b. If Oswald hadn’t killed Kennedy, someone else would

c. If Oswald didn't kill Kennedy, someone else would

d. If Oswald wasn’t killing Kennedy, someone else was

Are these more like (1) or more like (2)?

2. Two special uses of conditionals

von Fintel characterises indicatives and subjunctives as two of four categories, the others being ‘biscuit’ uses as in (7), and ‘premise’ uses as in (8):

(7) a. There are biscuits on the sideboard if you want some (Austin)

b. If you don’t mind my saying so, this is boring.

(8) If it is indeed that late, we should leave

(I’m not sure (8) is really a good example of what he’s tryng to get at here.) Unlike von Fintel, we will use the label ‘indicative’ in such a way as to encompass sentences like (7) and (8). It is not obvious that there is anything special about them from the point of view of semantics (as opposed to pragmatics). But until we have looked into the matter further we should be open to the idea that there is.

3. The material conditional analysis

A proposal: the proposition expressed by ’If ϕ, ψ’ (in a context) is identical to the one expressed by ‘Either not-ϕ, or ψ’ (in the same context). Or at least, these propositions are “equivalent” in the sense that each entails the other.

- We stipulatively define ‘$\supset$’ as the material conditional, i.e. ‘$\phi \supset \psi$’ means the same as ‘either not-ϕ or ψ’.

This theory could be applied to indicatives, counterfactuals, or both. Also, we could hold that within one of the two classes, some conditionals are material and others are not.

For counterfactuals, the material conditional analysis looks *insane*, and as far as I am aware it has no current defenders. It’s easy to come up with examples of counterfactuals with false antecedents that seem obviously false:

(9) If I had snapped my fingers I would have disappeared

By contrast, the claim that indicative conditionals are material conditionals has some distinguished defenders (although it is very much a minority view).

Why the contrast?

- Perhaps it isn't so easy to come up with counterexamples as glaring as (9)? (What do we think?)
- In the case of indicatives, the **or-to-if inference**—from 'ϕ or ψ' to 'if not ϕ, then ψ'—seems prima facie valid, whereas the inference from 'ϕ or ψ' to 'if it had been that not-ϕ, it would have been that ψ' looks atrocious in many cases. But if the or-to-if inference really is valid for indicative conditionals, it is very hard to avoid the concusion that they are material conditioals.

On the other hand, the view that indicatives are material conditionals faces many serious objections.

- Famous objections concern arguments that come out valid according to the view, but that look kind of weird: the so-called 'paradoxes of material implication'. First 'paradox': the argument from'not ϕ' to 'if ϕ then ψ'. Seond 'paradox': the argument from ψ to 'if ϕ then ψ'.
- One that I think is epsecially serious: often it seems appropriate to be *much less confident* that if ϕ, ψ than that $\phi \supset \psi$. For example: you should be very confident that either you will never win the lottery or you will be elected President (since you sould be very confident that you will never win the lottery). But you shouldn't be at all confident that if you win the lottery you will be elected President. (Right?)

We will return later to the debate about whether indicative conditonals are material conditionals.

Reading for next week:

- Lewis, *Counterfactuals*, chapter 1 (on NYU Classes)
- Bennett, chapter 10

Worlds, strict conditionals, and variably strict conditionals

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1. Background on propositions and possible worlds

Three important operations on propositions: conjunction (∧), disjunction (∨), negation (¬).

According to a view I call *Booleanism*, propositions form a *Boolean algebra* with respect to these operations. What this means: for any propositions p, q, r

(Commutativity)	(a) $p \wedge q = q \wedge p$	(b) $p \vee q = q \vee p$
(Distributivity)	(a) $p \wedge (q \vee r) = (p \wedge q) \vee (p \wedge r)$	(b) $p \vee (q \wedge r) = (p \vee q) \wedge (p \vee r)$
(Absorption)	(a) $p \wedge (q \vee \neg q) = p$	(b) $p \vee (q \wedge \neg q) = p$

It turns out to follow from these axioms that whenever the material conditional '$\phi \equiv \psi$' is a theorem of propositional logic, the proposition that ϕ = the proposition that ψ.

Some further definitions that make sense, given Booleanism:

p **entails** q, or for short '$p \leq q$', iff $p \wedge q = p$, or equivalently iff $p \vee q = q$. (Note that the above axioms entail that if $p \leq q$ and $q \leq p$ then $p=q$.)

p is the **greatest lower bound** of set S: p entails every member of S and is entailed by every other proposition that entails every member of S. (Note that $p \wedge q$ is the glb of {p,q}. The glb of S is also called the *conjunction* of the members of S.)

p is the **least upper bound** of set S: p is entailed by every member of S and entails every other proposition that is entailed by every member of S. (Note that $p \vee q$ is the lub of {p,q}. The lub of S is also called the *disjunction* of the members of S.)

$\bot$ is $p \wedge \neg p$ (for any p — note that the above axioms entail that $p \wedge \neg p = q \wedge \neg q$)

$\top$ is $p \vee \neg p$ (for any p — note that the above axioms entail that $p \vee \neg p = q \vee \neg q$)

A proposition a is an *atom* iff $a \neq \bot$ and for every b, if $b \leq a$, then $b=a$ or $b=\bot$.

A further hypothesis (that makes life very easy): the propositions form a *complete atomic* Boolean algebra. What this means:

(Completeness) Any set S of propositions has a greatest lower bound and a least upper bound.

(Atomicity) For every $p \neq \bot$, there is an atom a such that $a \leq p$.

Given these further hypotheses, there is a one-to-one correspondence between propositions and *sets of atoms*: each proposition corresponds to the set of atoms that entail it; each set of atoms corresponds to its least upper bound (disjunction). We can identify *worlds* with atoms.

Booleanism is a controversial view. However, it is much less controversial to hold that there is some relation of 'mutual entailment' on propositions, such that the Boolean axioms become true when '=' is interpreted to mean logical equivalence rather than identity. On this view, the set of all propositions does not form a Boolean algebra, but the set of *equivalence classes of propositions under mutual entailment* does form a Boolean algebra. And it still looks pretty plausible that *this* algebra is complete and atomic. If we think it is, we can identify "possible worlds" with the atoms of this algebra. Every proposition will correspond to a set of possible worlds—the set of atoms whose members entail it—and in cases where we don't care about the differences between mutually entailing propositions, we won't go wrong if we just equate the propositions with the corresponding sets of worlds.

- To make this more precise: say that a function *f* from propositions is *non-hyperintensional* iff, whenever p and q are mutually entailing, *f*(p) and *f*(q) are mutually entailing.

For many operations on propositions, it will be plausible to hold that applying the operation to logically equivalent inputs yields logically equivalent outputs. Those kinds of operations will be associated with functions from sets of worlds to sets of worlds; we get the set of worlds corresponding to the result of the operation by applying the function to the set of worlds corresponding to the input. If we only care about identifying the outputs of the operation up to mutual entailment, we will lose nothing by just looking at the corresponding function on sets of worlds.

2. Background on modal logic.

'Necessary' and 'possible' can mean many different things. Think of their meanings as *properties of propositions*, or the associated *functions from propositions to propositions*. We take it that these are non-hyperintensional.

- For each meaning of 'necessary' there is a corresponding meaning of 'possible', such that 'It is necessary that ϕ' is logically equivalent to 'it is not possible that not-ϕ', and 'It is possible that ϕ' is logically equivalent to 'It is not necessary that not-ϕ'. ('Duality').
- 'Necessarily (ϕ_1 and ϕ_2 and … ϕ_n)' is logically equivalent to 'Necessarily ϕ_1, and necessarily ϕ_2, and … and necessarily ϕ_n'. Similarly, 'Possibly (ϕ_1 or ϕ_2 or … ϕ_n)' is logically equivalent to 'Possibly ϕ_1, or possibly ϕ_2, or … or possibly ϕ_n'.
- This is also true for *infinite* conjunctions and disjunctions.

- So, the actions of the functions corresponding to by 'possibly' and 'necessarily' on *arbitrary* sets of worlds is completely pinned down once we are given the action of these functions on *singleton* sets of worlds. The result of applying the 'possibly' function to a set of worlds is just the union (disjunction) of the results of applying the operation to its singleton subsets.
- When the result of applying the possibility-function to {v} contains w, say that v is *accessible from* w, or '*w* sees *v*', sometimes written $w<v$. So we have the following:
    - '$\Diamond\phi$' is true at w iff ϕ is true at some world accessible from w
    - '$\Box\phi$' is true at w iff ϕ is true at every world accessible from w
- So: up to mutual entailment at least, a pair of corresponding interpretations of 'necessary' and 'possible' is fully specified by its corresponding accessibility relation.
- We can then go on to further categories modalities by looking at properties of their corresponding accessibility relations, such as reflexivity, symmetry, and transitivity. For example, reflexivity (every world is accessible from itself) corresponds to the claim that 'Necessarily ϕ' entails ϕ and ϕ entails 'Possibly ϕ'.

3. The strict conditional analysis of conditionals

Schematic proposal: the proposition expressed by 'If ϕ, ψ' is equivalent to the one expressed by 'Necessarily, either not-ϕ or ψ', *on some interpretation of "necessarily" or other*.

- In terms of accessibility: there is an accessibility relation such that 'If ϕ, ψ' is true at w iff ψ is true at every ϕ-world that is accessible from w.

The strict conditional analysis lets us avoid the objections to the material conditional analysis considered last week. The 'paradoxes of material implication' are no longer valid; and it can be rational to be much less confident in the truth of 'if ϕ, ψ' than in that of '$\phi\supset\psi$', for the same reason that it can be rational to be much less confident in the truth of 'necessarily ϕ' than in the truth of ϕ.

- Given that we are being so liberal about what counts as a possible interpretation of 'necessarily', the material conditional analysis is a *special case of* the strict conditional analysis—consider the "trivial modality" whose associated accessibility relation is the relation each world bears only to itself. This makes 'necessarily ϕ' equivalent to ϕ.
- If the strict conditional analysis is correct, there is plausibly quite a lot of vagueness and context-sensitivity associated with conditionals: different uses of 'if' will often correspond to different accessibility relations.
- Here are some suggestions for how it might go:

- for a typical use of indicative conditional, v is accessible from w iff v is compatible with what the speaker knows at w. (More generally, we may want to look not just at the speaker but at other contextually relevant people, and we may want to consider times other than the time of speech).
- for a typical use of a counterfactual conditional, v is accessible from w if v is compatible with (i) the laws of nature of w, and (ii) the history of w up to some contextually relevant time.

- From the point of view of a strict-conditionalist, it's plausible to think that certain uses of ordinary modals can pick up on the same accessibility relations that matter for conditionals. For indicatives, the relevant modals would be *epistemic* modals like 'Must' and 'might'/'may'; for counterfactuals, they would be so-called *circumstantial* modals, like 'had to' and 'could have'.

4. Lewis and Stalnaker's objections to the strict conditional account

The following three controversial argument-forms are all valid on the strict condiitonal approach:

> **Antecedent Strengthening:** If A, C; therefore if A and B, C.
>
> **Transitivity:** If A, B; if B, C; therefore if A, C.
>
> **Contraposition:** If A, B; therefore if not B, not A.

Influentially, Lewis and Stalnaker offered counterexamples to these inference rules—at least for counterfactuals—and used them to argue for an alternative to the strict conditional anlaysis.

Counterexamples to antecedent strengthening for counterfactuals:

> If you walked on the grass it would be fine. ? Therefore, if you and everyone else in the building walked in on the grass, it would be fine.

Counterexample to transitivity for counterfactuals (Stalnaker)

> (Stalnaker) If Hoover had been Russian, he would have been a communist. If Hoover had been a communist he would have been a traitor. ? Therefore, if Hoover had been Russian, he would have been a traitor.

Counterexample to contraposition for counterfactuals:

> If I had talked to Mary, I wouldn't have told her anything important. ? Therefore, if I had told Mary something important, I wouldn't have talked to her.

Variably strict conditionals

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1. Lewis's analysis, expressed using accessibility and closeness

Following Lewis, let's use 'A $\Box\!\!\rightarrow$ B' to symbolise 'If it were the case that A, it would be the case that B'. Let's also say 'w is an A-world' to mean 'A is true at w', i.e. 'The proposition expressed by A is true at w'.

New gizmo: a three-place *closeness relation* among possible worlds. Write as '$u \leq_w v$', pronounce as 'u is at least as close to w as v'.

Closeness Analysis: A $\Box\!\!\rightarrow$ B is true at w iff either (a) no A-world is accessible from w, or (b) there is an A-world v accessible from w, such B is true at every A-world u such that $u \leq_w v$ and u is accessible from w.

- Lewis only defends this analysis for counterfactuals; he sets indicative conditionals aside. One *could* however defend an analysis of this form for both indicative and counterfactual conditionals; Stalnaker did this at one point. If you do this, you'll need to explain the difference between indicatives and counterfactuals by appeal to some difference in the operative closeness relation and/or accessibility relation.
- When condition (a) holds we say that A $\Box\!\!\rightarrow$ B is *vacuously true*; when (b) holds, we say that it is *non-vacuously true*.

2. Closeness and similarity

The Closeness Analysis doesn't say what closeness is: it's a black box that could be filled in any number of different ways. But Lewis has a proposal: for u to be closer to w than v is for u to be *more similar* to w, in relevant respects, than v is.

Objection:

> The counterfactual "If Nixon had pressed the button there would have been a nuclear holocaust" is true or can be imagined to be so. Now suppose that there never will be a nuclear holocaust. Then that counterfactual is, on Lewis's analysis, very likely false. For given any world in which antecedent and consequent are both true it will be easier to imagine a closer world in which the antecedent is true and the consequent false. For we need only imagine a change that prevents the holocaust but that does not require such great divergence from actuality. (Kit Fine, Critical Notice of *Counterfactuals*)

Lewis's response:

> It is all too easy to make offhand similarity judgments and then assume that they will do for all purposes. But if we respect the extreme shiftiness and context-dependence of similarity, we will not set much store by offhand judgments. We will be prepared to distinguish between the similarity relations that guide our offhand explicit judgments and those that govern our counterfactuals in various contexts….
>
> The presence or absence of a nuclear holocaust surely does contribute with overwhelming weight to some prominent similarity relations…. But the relation that governs the counterfactual may not be one of these. It may nevertheless be a relation of overall similarity … because it is a resultant, under some system of weights or priorities, of a multitude of relations of similarity in particular respects. (Lewis, 'Counterfactual Dependence and Time's Arrow', pp. 42–3)

3. Accessibility

The notion of "accessibility" does much less work in the Closeness Analysis than in the Strict Conditional Analysis. To see this, consider the following hypothesis:

Universality Hypothesis: every world is accessible from every world

Given the Strict Condiitonal Analysis, adopting this would have the crazy result that 'If A, B' is true only when $A \wedge \neg B$ is metaphysically impossible (true at no worlds). By contrast, the Universality Hypothesis can be (and often is) combined with the Closeness Analysis without leading to any obviously problematic judgments about the truth-values of particular conditionals. If we do this, we can simplify the statement of the Closeness Analysis slightly by dropping all mention of accessibility.

4. The Basic Properties and valid argument-schemes

In order to secure a reasonable logic for conditionals from the Closeness Analysis, we will need to assume that the following "Basic Properties" of the accessibility and closeness relations:

> ***Accessibility is reflexive:*** every world is accessible from itself.
>
> ***Transitivity***: if $u_1 \leq_w u_2$ and $u_2 \leq_w u_3$ then $u_1 \leq_w u_3$.
>
> ***Connectedness:*** for any u, v, and w, either $u \leq_w v$ or $v \leq_w u$.
>
> ***Weak centering***: $w \leq_w v$ for any w and v.

(Note that the connectedness property entails that $v \leq_w v$ for every w and v.)

Provided we require the closeness and accessibility relations to have the Basic Properties, the following argument-schemas will be *valid* (in the sense that no matter how we interpret the sentence-letters, the conclusion is true at every world where all the premises are true)

ID (Identity): $\vdash A \mathrel{\Box\!\!\rightarrow} A$

MP (Modus Ponens): $A \mathrel{\Box\!\!\rightarrow} B,\ A \vdash B$

CC (Agglomeration): $A \mathrel{\Box\!\!\rightarrow} B, A \mathrel{\Box\!\!\rightarrow} C \vdash A \mathrel{\Box\!\!\rightarrow} (B \wedge C)$

CA (Disjunction): $A \mathrel{\Box\!\!\rightarrow} C, B \mathrel{\Box\!\!\rightarrow} C \vdash (A \vee B) \mathrel{\Box\!\!\rightarrow} C$

CSO (Restricted Transitivity): $A \mathrel{\Box\!\!\rightarrow} B, B \mathrel{\Box\!\!\rightarrow} A, A \mathrel{\Box\!\!\rightarrow} C \vdash B \mathrel{\Box\!\!\rightarrow} C$

CV (Limited Antecedent Strengthening): $A \mathrel{\Box\!\!\rightarrow} C \vdash (A \wedge B \mathrel{\Box\!\!\rightarrow} C) \vee (A \mathrel{\Box\!\!\rightarrow} \neg B)$

Example 1: To show that MP is valid, suppose that $A \mathrel{\Box\!\!\rightarrow} B$ and A are both true at some world w. By the reflexivity of accessibility, w is accessible from itself, so A is true at some world accessible from w, so $A \mathrel{\Box\!\!\rightarrow} B$ is not vacuously true at w. So there is an A-world v accessible from w such that whenever $u \leq_w v$ and u is an A-world accessible from w, B is true at u. By Weak Centering, $w \leq_w v$, and by the reflexivity of accessibility, w is an A-world accessible from w. So B is true at w.

Example 2: To show that CC is valid, suppose that $A \mathrel{\Box\!\!\rightarrow} B$ and $A \mathrel{\Box\!\!\rightarrow} C$ are both true at some world w. If there is no A-world accessible from w, then $A \mathrel{\Box\!\!\rightarrow} (B \wedge C)$ is vacuously true at w. Otherwise, there are A-worlds v_1 and v_2 accessible from w such that for any A-world u accessible from w, B is true at u if $u \leq_w v_1$, and C is true at u if $u \leq_w v_2$. By Connectedness, either $v_1 \leq_w v_2$ or $v_2 \leq_w v_1$. Suppose the former. And suppose that u is an A-world accessible from w such that $u \leq_w v_1$. Then B is true at u; also $u \leq_w v_2$ by the Transitivity Property, so C is true at u. Therefore $B \wedge C$ is true at u. Since this holds for every u, we can conclude that v_1 is an accessible A-world such that $A \wedge B$ is true at every A-world that is at least as close to w as it. Similarly, if $v_2 \leq_w v_1$, v_2 has the same feature. So either way, $A \mathrel{\Box\!\!\rightarrow} (B \wedge C)$ is true at w.

We can get further argument-schemas to come out valid by imposing more requirements on the accessibility and/or closeness relations.

One important example: if we require the closeness relation to have the

Strong centering property: For any w and v, $v \leq_w w$ iff $v=w$.

the following schema will be valid

CS (And-to-if): $A, B \vdash A \mathrel{\Box\!\!\rightarrow} B$

5. Invalidities

Even given the Basic Properties, Antecedent Strengthening is no longer valid: $A \mathrel{\Box\!\!\rightarrow} C$ can be true at a world without $(A \wedge B) \mathrel{\Box\!\!\rightarrow} C$ being true at that world.

- We can find a countermodel to this with just two worlds w_1 and w_2, each accessible from the other, such that $w_2 \leq_{w1} w_1$ is false. Let A be true at both worlds, B be true only at w_2, and C be true only at w_1; then A $\Box\!\!\rightarrow$ C is true at w_1 although $(A \wedge B) \Box\!\!\rightarrow C$ is false at w_1.

Similarly for Transitivity and Contraposition.

6. Reformulation in terms of spheres

(D1) Let a **sphere around *w*** be a set S of worlds, all accessible from *w*, such that $u \in S$ whenever u is accessible from w, $u \leq_w v$, and $v \in S$.[1]

Notice that given the Basic Properties this definition of 'sphere' entails that

(a) Nestedness: if S_1 and S_2 are both spheres around w, then either $S_1 \subseteq S_2$ or $S_2 \subseteq S_1$.

(b) The union of any set of spheres around w is a sphere around w

(c) The intersection of any nonempty set of spheres of around w is a sphere around w.

(d) Weak centering: Every nonempty sphere around w contains w and there is at least one nonempty sphere around w.

Given (D1), the properties of $\leq$, and the the Closeness Analysis, we can derive

Spheres Analysis: A $\Box\!\!\rightarrow$ C is true at w iff either (a) no sphere around w contains an A-world, or (b) some sphere around w contains an A-world and is such that every A-world in it is a B-world.

Alternatively, we could take the notion of a sphere around a world as primitive, and define accessibility and closeness in terms of it, as follows:

(D2) v is accessible from w $=_{df}$ v belongs to some sphere around w.

(D3) $u \leq_w v =_{df}$ every sphere around w that contains v also contains u.

Note that given the assumption that spheres obey conditions (a)–(d), this definition entails that accessibility and closeness have the Basic Properties.

Given (D2) and (D3), conditions (a)–(d) on spheres, and the Spheres Analysis, we can derive the Closeness Analysis. Hence the two formulations are completely equivalent.

[1] The version of this handout distributed in class erroneously omitted the requirement that u be accessible from w. Thanks to Alex Poste for spotting the error.

Strengthenings of Lewis's logic

Cian Dorr

19 September 2017

1. Recap

Closeness Analysis: A $\Box\!\!\rightarrow$ B is true at w iff either (a) no A-world is accessible from w, or (b) there is an A-world v accessible from w, such that B is true at every A-world u such that u $\leq_w$ v and u is accessible from w.

"Basic Properties" of the accessibility and closeness relations:

Accessibility is reflexive: every world is accessible from itself.

Transitivity: if $u_1 \leq_w u_2$ and $u_2 \leq_w u_3$ then $u_1 \leq_w u_3$.

Connectedness: for any u, v, and w, either u $\leq_w$ v or v $\leq_w$ u.

Weak centering: w $\leq_w$ v for any w and v.

Valid argument schemas:

ID (Identity): $\vdash A \Box\!\!\rightarrow A$

MP (Modus Ponens): $A \Box\!\!\rightarrow B,\ A \vdash B$

CC (Conjunction): $A \Box\!\!\rightarrow B, A \Box\!\!\rightarrow C \vdash A \Box\!\!\rightarrow (B \wedge C)$

CA (Disjunction): $A \Box\!\!\rightarrow C, B \Box\!\!\rightarrow C \vdash (A \vee B) \Box\!\!\rightarrow C$

CSO (Weakened Transitivity): $A \Box\!\!\rightarrow B, B \Box\!\!\rightarrow A, A \Box\!\!\rightarrow C \vdash B \Box\!\!\rightarrow C$

CV (Limited Antecedent Strengthening): $A \Box\!\!\rightarrow C, \neg(A \Box\!\!\rightarrow \neg B) \vdash (A \wedge B) \Box\!\!\rightarrow C$

2. Counterfactuals and metaphysical necessity

Two other schemas that are valid on Lewis's system, when $\Box$ is interpreted as metaphysical necessity

Entailment in the consequent:

$\Box(B \supset C), A \Box\!\!\rightarrow B \vdash A \Box\!\!\rightarrow C$

Equivalent antecedents:

$\Box(A \equiv B), A \Box\!\!\rightarrow C \vdash B \Box\!\!\rightarrow C$

Since valid argument-schemas are necessarily truth-preserving, these correspond to two useful *rules of proof*:

Deduction in the consequent:

Whenever $B \vdash C$, $A \Box\!\!\rightarrow B \vdash A \Box\!\!\rightarrow C$

Substitutivity in the antecedent:

Whenever A⊢B and B⊢A, A □→ C ⊢ B □→ C

Note that these are not *argument-schemas*, but *metarules* which tell us that some arguments are valid conditional on others being valid.

3. The Limit Assumption and the Closest-Worlds Analysis

Consider the following possible add-on principle about closeness:

The Limit Assumption: for any non-empty set of worlds S, there is some u∈S such that $u \leq_w v$ for every v∈S. (In other words.: each $\leq_w$ is *well-founded.*)

Note that if we make the convenient (but false!) assumption that there are only finitely many possible worlds, the Limit Assumption will follow automatically from the Connectedness and Transitivity properties. (Prove this!)

Given the Well-foundedness Property we can restate the Closeness Analysis more simply, as follows:

Closest-Worlds Analysis: A □→ B is true at w iff B is true at every A-world accessible from w that is at least as close to w as any A-world accessible from w. (B is true at every *maximatlly close* accessible A-world.)

Proof: Left to right: Suppose A □→ B is true at w. And suppose that v′ is a maximatlly close accessible A-world to w. Since condition (a) of the Closeness Analysis does not obtain, condition (b) must obtain: there is an A-world v accessible from w such that such that B is true at every A-world u such that $u \leq_w v$. But since v′ is a closest accessible A-world to w, $v' \leq_w v$, so B is true at v′. Since v′ was arbitrary, we can conclude that B is true at every closest accessible A-world to w.

Right-to-left: Suppose B is true at every closest accessible A-world to w. If the set of A-worlds accessible from w is empty, then condition (a) of the Closeness Analysis obtains, so A □→ B is true at w. Otherwise, by the Well-Foundedness Property, there is a maximally close accessible A-world to w. Let v be any such world, let u be any accessible A-world such that $u \leq_w v$, and let v′ be any other accessible A-world. Since v is a maximally close accessible A-world to w, $v \leq_w v'$. So by the Transitivity Property, $u \leq_w v'$. Since v′ was arbitrary, we can conclude that u is also a maximally close accessible A-world to w, and so u is a B-world, by our hypothesis. Since u was arbitrary, we can conclude that B is true at every accessible A-world u such that $u \leq_w v$; thus condition (b) of the Closeness Analysis obtains, and so again in this case, A □→ B is true at w.

4. The Limit Assumption and Infinite Conjunction

Infinite Conjunction: If a C is a collection of propositions each of which would have been true if A had been true, then if A had been true, all the members of C would have been true.

This is *not valid* in Lewis's system without the Limit Assumption. For each accessible A-world w, let P_w be a proposition true at all and only the A-worlds that are at least as close to the actual world as w; and let the collection C contain P_w for every A-world w. Suppose that althought there are accessible A-worlds, there is no closest one (to the actual world). Then there is no accessible A-world where all the members of C are true, so 'If A had been true, all the members of C would have been true' is false. Nevertheless, each $P_w \in C$ is such that 'If A had been true, P_w would have been true' is true, since P_w is true at the A-world w and all closer A-worlds to it.

5. Lewis against the Limit Assumption

Lewis's Line: for every world w where the line is more than 1 inch long, there is a world w′ that is closer to actuality than w, where the line is still more than 1 inch long but shorter than it is in w.

Objection to this argument: If the closeness relation worked like this, then given the Closeness Analysis, all sentences of the form 'If Lewis's Line had been more than 1 inch long it would have been less than $1+n$ inches long', would be true, for any positive n. But many such sentences seem quite implausible, e.g. where $n = 0.001$.

Pollock's claim: Infinite Conjunction is just as plausible as Conjunction.

- Upshot: if we want to keep the Closeness Analysis, we should think that the closeness relation (for any context) has to be one that obeys the Limit Assumption.
- Question: does this require giving up the idea that closeness is a kind of comparative similarity relation?

6. Other possible additions to the system

Property of closeness/accessibility	*Valid argument schema*
***Strong Centering**:* $v \leq_w w$ iff $v=w$	**And-to-if:** $A, B \vdash A \mathrel{\Box\!\!\to} B$
Antisymmetry: If $u \leq_w v$ and $v \leq_w u$ then $u=v$	**CEM (Conditional Excluded Middle)**: $\vdash (A \mathrel{\Box\!\!\to} B) \vee (A \mathrel{\Box\!\!\to} \neg B)$
Boringness: If u and v are both accessible from w, then $u \leq_w v$.	**AS (Antecedent Strengthening)**: $A \mathrel{\Box\!\!\to} C \vdash (A \wedge B) \mathrel{\Box\!\!\to} C$
Triviality: If u is accessible from w then u=w.	**Or-to-if:** $A \vee B \vdash A \mathrel{\Box\!\!\to} \neg B$

Counterfactuals, Laws, and History

Topics in Language and Mind
21 September 2017

1. Holding History Fixed

Claim: usually, when we evaluate an ordinary counterfactual 'If it were that A it would be that C', where A is concerned with a particular interval of time, we tacitly "hold fixed" a very broad range of propositions about history before that interval, in the sense that we assume that if these propositions are in fact true, they would also have been true if A had been true.

Illustrations:

(1) If I had done such-and-such, I would have revenged a wrong done me last year.
(2) If John had skipped breakfast today, that would have been the first time he did so in months.
(3) If this video were downloaded 10,000 times, that would set a new record for us.

However, as Lewis recognises, there are some circumstances where we do not reason like this. Contrast:

(4) If I had jumped out of that tenth-floor window, I would have been killed.
(5) If I had jumped out of that tenth-floor window, that would have to have been because someone had a safety net in place. (Jackson)

These two sentences naturally suggest two *different contexts.* If they were both true in the same context,

(6) If I had jumped out of that tenth-floor window, I would have been killed and there would have been a safety net in place.

would also have to be true in that context; but this doesn't seem to be true in any normal context. Lewis calls the kind of context evoked by (5) a "backtracking context", and the kind of context evoked by (4) a "standard context".

Lewis's project: give an analysis that captures the truth conditions of 'If it were that A, it would be that C' in a *standard* context.

- Note that one might think there was further important context-sensitivity *within* the set of standard contexts. (Come back to this.)

2. An account not in terms of closeness

Assuming we accept Lewis's logic for counterfactuals, there is good reason to think that any possible reading a counterfactual *can* be captured by some combination of a closeness relation and an accessibility relation. But this does not mean we are *forced* to statethe truth conditions in this way! Lewis begins with an analysis, 'Analysis 1', that is not expressed in terms of closeness / acessibility:

> ANALYSIS 1. Consider a counterfactual "If it were that A, then it would be that C", where A is entirely about affairs in a stretch of time t_A. Consider all those possible worlds w such that:
>
> > (1) A is true at w;
> > (2) w is exactly like our actual world at all times before a transition period beginning shortly before t_A;
> > (3) w conforms to the actual laws of nature at all times after t_A, and
> > (4) during t_A and the preceding transition period, w differs no more from our actual world than it must to permit A to hold.
>
> The counterfactual is true if and only if C holds at every such world w.

Lewis thinks that Analysis generally assigns the right truth values to counterfactuals (understood relate to a standard context). His complaints are as follows:

- It is too weak - it is just silent about the truth conditions of counterfactuals whose antecedents are not entirely about affairs in a stretch of time.
- It incorrectly rules out backwards counterfactual dependence (outside the 'transition period') as *absolutely* impossible, even under weird laws of nature.

Further reasons to continue the inquiry: Analysis 1 does not fit into the general closenes framework (or any other general framework that deals with all contexts, not just the standard ones); and it leaves us in need of an explanation of why we have *this* concept, rather than a different concept in which the roles of 'past' and 'future' are reversed.

3. Determinism

Determinism: all truths follow with metaphysical necessity from (i) the laws of nature, together with (ii) the complete intrinsic truth about any initial segment of the history of the universe.

4. A controversial feature that Analysis 1 shares with Lewis's final view: miracles

Suppose determinism is true. Both of the following claims seem tempting:

> *Laws*: If you hadn't come to class today, every actual law of nature would have been true.
>
> *Past*: If you hadn't come to class today, the intrinsic character of the first billion-year stretch of the history of the universe since the Big Bang would have been exactly the same.

But assuming determinism, the actual laws plus the intrinsic truth about that interval jointly entail that you do come to class today. Given the principle that P > Q entails P >

R whenever P is metaphysically possible and Q entails R, this means that if we accept both *Laws* and *Past* we have to say that if you hadn't come to class today you would have come to class today, which is absurd. So, either *Laws* or *Past* have to be given up.

Lewis gives up *Laws:* he thinks that assuming determinism is true, if things had been different at one time, a 'miracle' would have occurred shortly before that time.

> NB: That does not mean that false propositions can be laws. The miracle is inconsistent with the *actual* laws, not with anything that would have been a law if you hadn't come to class.

An alternative approach: keep *Laws* and reject *Past*, even for ordinary ('non-backtracking' contexts).

> Does that mean just rejecting the asymmetry of counterfactuals altogether? Not necessarily. We could say that if things had been different at t, the course of history before t would have been only *slightly* different, whereas the course of history after t would have been significantly different.
>
> Lewis worries about this: 'Indeed, it is hard to imagine how two deterministic worlds anything like ours could possibly remain just a little bit different for very long. There are altogether too many opportunities for little differences to give rise to big differences.' (45).

- In 'Against Counterfactual Miracles', section 4, I argue that there is nothing to worry about here.

Q: how could Analysis 1 be revised to fit with the law-preserving approach?

Q: How could Lewis's final analysis be revised to fit with the law-preserving approach?`

5. **Another controversial feature that Analysis 1 shares with Lewis's final view: the "transition period"**

Lewis does not think that if things had been intrinsically different at *t* the *entire* past of *t* would have been *exactly* the same—he allows for a 'transition period'.

> What motivates this?

Lewis's closeness relation
Topics in Language and Mind
26 September 2017

1. Recap

According to Lewis the following analysis yields correct verdicts about the truth values of counterfactuals (assuming determinism and the absence of time travel, etc.):

> ANALYSIS 1. Consider a counterfactual "If it were that A, then it would be that C", where A is entirely about affairs in a stretch of time t_A. Consider all those possible worlds w such that:
>
> > (1) A is true at w;
> > (2) w is exactly like our actual world at all times before a transition period beginning shortly before t_A;
> > (3) w conforms to the actual laws of nature at all times after t_A, and
> > (4) during t_A and the preceding transition period, w differs no more from our actual world than it must to permit A to hold.
>
> The counterfactual is true if and only if C holds at every such world w.

(Note however that the predictions of Analysis 1 are not very specific since we are not told holw long the "transition period" ought to be.)

Two controversial features of this view:

- Given determinism, laws of nature are counterfactually super-fragile: if A had been true (for any A entirely about a stretch of time with a long enough time before it), the <u>actual</u> laws of nature would have been false.
- We are allowed a "transition period": Lewis does not say that if A had been true *t* things would have been exactly the same at *all* times before t_A. Motivation: avoiding what Bennett calls 'bumps'.

(1) If the Germans had reached Moscow in August they would have had (very easily) captured the city
(2) If I had begun driving to Princeton at noon I would have teleported from my kitchen into the car
(3) If I had begun driving to Princeton at noon I would have been in grave danger of crashing (due to the surprise and disorientation of teleportation)…

2. Lewis's account of the closeness relation (for standard contexts)

(L1) It is of the first importance to avoid big, widespread, diverse violations of law.

(L2) It is of the second importance to maximize the spatio-temporal region throughout which perfect match of particular fact prevails.

(L3) It is of the third importance to avoid even small, localized, simple violation of law.

(L4) It is of little or no importance to secure approximate similarity of particular fact, even in matters that concern us greatly.

Lewis's claim: Assume that the actual world is deterministic, and that Nixon's button is wired straight up, at t, to the nuclear missile launcher. Then, by this measure of similarity, the most similar worlds to the actual world where Nixon presses the button at t are ones which match actuality perfectly up to shortly up to t, then diverge via a small miracle (i.e. a violation of the *actual* laws of nature), then proceed according to the actual laws of nature, in such a way that the nuclear holocaust ensues.

Let w_1 be such a world, and let w_0 be the actual world where Nixon does not press the button. What motivates the structure of Lewis's analysis of closeness is the need to get w_1 to come out closer to w_0 than any of the following competitors, in all of which Nixon presses the button.

- w_2—a world where the entire past is different and there are no miracles at all.
- w_3—a world where the past up to shortly before t is the same, whereupon a small miracle leads to Nixon's pushing the button, after which another small miracle leads to the signal's not getting all the way to the launcher.
- w_4—a world where the past up to shortly before t is the same, whereupon a small miracle leads to Nixon's pushing the button, whereupon a *big* miracle wipes out all of the traces of his having done so (fingerprints, click on tape, light signals shooting out into space…)

If there could be a world w_5 where a small miracle leads to the button being pushed and a second, later small miracle wipes out all the traces, that would come out closer than w_1 to w_0. But Lewis denies that there is any such world.

3. How is this time-symmetric analysis supposed to get time-asymmetric results?

The following temporally asymmetric fact is supposed to be true of our world: it takes only a small miracle to diverge from it, but it takes a big miracle to converge to it. I.e. any worlds that are exactly like it at all times after a certain time t but not beforehand contain a large, widespread violation of the laws of the actual world.

Is this true?

9. Pollock's objection to Lewis

(4) If my coat had been stolen last year, it would have been stolen on December 31st.

Tendentious objection: Lewis's theory says (4) is true, but it’s false.

A better objection: Lewis’s theory says (4) is knowably true, but it’s not knowably true.

Context-sensitivity, Pollock's coat, and strictism

Topics in Language and Mind
26 September 2017

1. Pollock's objection to Lewis

(1) If my coat had been stolen last year, it would have been stolen on December 31st.

Tendentious objection: Lewis's theory says (1) is true, but it's false.

A better objection: Lewis's theory says (1) is knowably true, but it's not knowably true.

2. Not just Lewis's problem

Background: a big storm was raging all day on Saturday; it died down late on Saturday night, long after I went to bed. Happily, my favourite tree managed survived the storm.

(2) If this tree had been in pieces on Sunday morning, I would have been too sad to eat breakfast

<u>In the context evoked by (2)</u>, it looks like (3) has to be true:

(3) If the tree had been in pieces on Sunday morning, it would have been intact on Saturday at bedtime.

But also (4) and (5) look obviously true:

(4) If the tree had been in pieces on Sunday morning, it would have been destroyed on Saturday.

(5) If the tree had been destroyed on Saturday, it would have been in pieces on Sunday morning.

Now recall the argument-schema CSO ('Weakened Transitivity'):

CSO: $A > B, B > A, A > C \vdash B > C$

Applying CSO to (3)–(5) gives us

(6) If the tree had been destroyed on Saturday, it would have been intact on Saturday at bedtime

This seems bad in the same way as Pollock's coat example. Similarly, applying CSO to (2), (4), (5) yields the similarly problematic

(7) If this tree had been destroyed on Saturday, I would have been too sad to eat breakfast on Sunday

3. Three possible solutions to this problem

(i) Bite the bullet: say that (6) and (7) are just true, and try to explain away our resistance to them.

(ii) Deny CSO.

(iii) Appeal to context-sensitivity. Basic strategy: in the context naturally evoked by (2), all of the above sentences express propositions that we know are true. But in the context naturally evoked by (6) and (7), none of (2), (3), (6), and (7) express propositions that we know to be true.

4. Some support for CSO

We can derive CSO from two other rules whose appeal is more immediately apparent:

RCV ('Very Limited Antecedent Strenghening'): $A > B, A > C \vdash (A \wedge B) > C$

CT ('Cumulative Transitivity'): $A > B, (A \wedge B) > C \vdash A > C$

Here is the derivation:

1.	$A > B$	Premise
2.	$B > A$	Premise
3.	$A > C$	Premise
4.	$(A \wedge B) > C$	1, 3, RCV
5.	$(B \wedge A) > C$	4, Substitutivity
6.	$B > C$	2, 5, CT

We can also derive CSO from RCV together with CA, CC, and ID, thus:

1.	$A > B$	Premise
2.	$B > A$	Premise
3.	$A > C$	Premise
4.	$(A \wedge B) > C$	1, 3, RCV
5.	$(A \wedge B) > (\neg A \vee C)$	4, Weakening
6.	$(\neg A \wedge B) > (\neg A \wedge B)$	ID
7.	$(\neg A \wedge B) > (\neg A \vee C)$	6, Weakening
8.	$((A \wedge B) \vee (\neg A \wedge B)) > (\neg A \vee C)$	7, CA
9.	$B > (\neg A \vee C)$	8, Substitutivity
10.	$B > (A \wedge (\neg A \vee C))$	2, 9, CC
11.	$B > C$	10, Weakening

5. Developing the contextualist response

Basic thought: which closeness and/or accessibility relation is operative in a context depends on which times are relevant in that context. Different antecedents tend to favour different resolutions of context-sensitivity, by making different times relevant. In the context of (2), Saturday daytime is not relevant, but in the context of (6) and (7), it is.

- An implementation using closeness: match as regards times before the contextually relevant period contributes to closeness; match as regards other times does not.
- An implementation using accessibility: the operative accessibility relation holds between two worlds only when they match as regards history before the contextually relevant time.
    - This could be combined with a strict conditional analysis or the closest-worlds analysis.

Back to strictness?
Topics in Language and Mind
3 October 2017

1. The case of 'every'

A standard picture: 'every' is context-sensitive; the contribution of context may be represented by a property D, the 'domain restrictor'. 'Every NP VP' is true in a context where the domain restrictor is D iff there is nothing that has D and the property expressed by the NP while lacking the property expressed by the VP.

Predictions: the following argument-schemes are valid in the sense that they preserve truth when the context (including the interpretation of 'every'):

> 'Antecedent strengthening': Every F is a G $\vdash$ Every F which is H is a G
>
> 'Transitivity': Every F is a G, Every G is a H $\vdash$ Every F is a H
>
> 'Contraposition': Every F is a G $\vdash$ Every non-G is a non-F

Counterexamples?

> ? Every philosopher went to the Dean's reception $\vdash$ Every philosopher at the University of Hong Kong went to the Dean's reception
>
> ? Every philosopher went to the Dean's reception, Everyone with an office in Building 9 in the University of Hong Kong is a philosopher $\vdash$ Everyone with an office in Building 9 in the University of Hong Kong went to the Dean's reception
>
> ? Every student who read that book didn't read it carefully $\nvdash$ Every student who read that book carefully didn't read that book

A standard response: these are not counterexamples: on the interpretations on which they might appear to be counterexamples, the context—in particular, the intepretation of 'Every'—is not being held fixed.

Challenge: explain why context tends to change like this

2. Presupposition and how it helps

Basic idea: in addition to their main (assertive) role, some sentences also signal that certain things are *to be taken for granted*. When a sentence carries a presupposition, its negation, and the corresponding yes/no question carry the same presupposition.

> John stopped smoking / John didn't stop smoking / Did John stop smoking?
>
> > $\rightarrow$ John used to smoke

Mary realised she was speeding / Mary didn't realise she was speeding / Did Mary realise she was speeding

→ Mary was speeding

I met my daughter / I didn't meet my daughter / Did I meet my daughter?

→ I have a daughter

More controversially:

Every F is a G / Not every F is a G / Is every F a G?

→ Something is an F

If it were that P it would be that Q / It is not the case that if it were that P it would be that Q / If it were that P would it be that Q?

→ It might have been that P

(= Some P-world is accessible [from the actual world])

If P, Q / It is not the case that if P, Q / If P, Q?

→ It might be that P

Idea: presuppositions help us fix the interpretations of context-sensitive expressions. When someone utters a sentence that carries a presupposition, we look for an interpretation on which the presupposition is really obvious enough to be *properly* taken for granted. This often trumps the pressure to interpret repeated occurrences of the same word uniformly across a discourse.

3. Aside: a contrast between domain expansion and domain contraction

The presupposition effect often creates pressure for domains to *expand* over the course of a discourse. Getting them to *contract* is harder.

Every philosopher was at the meeting. We discussed the fact that some philosophers in Hong Kong will be visiting next month.

At the meeting, we discussed the fact that some philosophers in Hong Kong will be visiting next month. ? Every philosopher was there.

4. A warning about the truth-value-gap theory of presupposition

5. Dynamic meanings and the role of similarity

Von Fintel's further idea: the process of adjustment of the accessibility parameter is "automated". If the currently operative accessibility relation is such that some worlds don't see any P-worlds, and someone utters 'If it were that P, it would be the case that Q', the current accessibility relation will expand just enough such that every world sees the most similar P-worlds to it, and all worlds at least as similar to it as they are.

- Note that this kind of mechanical evolution would not be plausible for 'every'. Is it well-motivated for 'if'?

Von Fintel's technical implementation of the further idea:

There are things called 'contexts', which we can treat (simplifying) as pairs of an accessibility relation and a closeness relation. In addition to having truth conditions $[\![\phi]\!]^{f,\leq}$ relative to every context, a sentence also has a dyamic meaning $|\phi|^{\leq}$ relative to a closeness relation $\leq$, which is a function from accessibility relations to accessibility relations.

$\phi>\psi$ is 'true in' a context $\langle f, \leq\rangle$ at w iff all the ϕ-worlds in $|\phi>\psi|^{\leq}(f)$ are ψ-worlds.

Reverse Sobel sequences, and a first glimpse of probabilities

Topics in Language and Mind
3 October 2017

1. An argument <u>for</u> the strict conditional approach: reverse Sobel sequences

Von Fintel and others hear a contrast between the different orders.

(1a) If the USA thr new its weapons into the sea tomorrow, there would be war; but if all the nuclear powers threw their weapons into the sea tomorrow, there would be peace.

(1b) ? If all the nuclear powers threw their weapons into the sea tomorrow, there would be peace; but if the USA threw its weapons into the sea tomorrow, there would be war.

(2a) If Sophie had gone to the parade, she would have seen Pedro. But if Sophie had gone to the parade and been stuck behind a tall person, she would not have seen Pedro.

(2b) ? If Sophie had gone to the parade and been stuck behind a tall person, she would not have seen Pedro. But if Sophie had gone to the parade, she would have seen Pedro.

Von Fintel's idea: explain the contrast by reference to the idea that quantifier domains can easily expand but not easily contract. So the pairs are analogous to the following pair using 'every':

(3a) Every philosopher is at the Dean's reception in the Silver Center. But every Chinese philosopher is at a conference in Beijing.

(3b) ? Every Chinese philosopher is at a conference in Beijing. But every philosopher is at the Dean's reception in the Silver Center.

- How bad are (1b) and (2b)? As bad as (3b)?
- Is there any similar contrast between (4a) and (4b)?

(4a) The closest town with a post-office is an hour away. But the closest town with a post-office and a hospital is five hours away.

(4b) The closest town with a post-office and a hospital is five hours away. But the closest town with a post-office is an hour away

2. Moss's epistemic diagnosis

Moss's thought: sometimes, a sentence A makes salient certain hypotheses whch we can't rule out with absolute certainty. Once such a hypothesis is salient, we are unwilling to flatly assert things which are inconsistent with it, or which are improbable given it, even when we are very confident that those things are true, and would have happily asserted them in a setting where the hypothesis was not salient.

(5a) That animal was born with stripes.
(5b) But cleverly disguised mules are not born with stripes.
(6a) Cleverly disguised mules are not born with stripes.
(6b) #But that animal was born with stripes.

According to Moss, 'If Sophie had gone to the parade and been stuck behind a tall person, she would not have seen Pedro' makes salient the following hypothesis, which can't be ruled out with certainty: it's not the case that if Sophie had gone to the parade, she would not have been stuck behind a tall person.

- According to Lewis, this hypothesis can be stated more easily as follows: if Sophie had gone to the parade, she might have been stuck behind a tall person.
- According to Stalnaker, the hypothesis entails the following: if Sophie had gone to the parade, she would have been stuck behind a tall person.
- Note that this hypothesis is not *inconsistent* with the claim that if Sophie had gone to the parade she would have seen Pedro. But that claim is certainly *improbable* given the hypothesis, which is all Moss needs.

It's not crucial that the sentence that does the salient-making is itself a counterfactual:

(7a) Do you remember when Kate got stuck behind a tall person and missed seeing Pedro in her first baseball parade?
(7b) ? But if Sophie had gone to the New York Mets parade, she would have seen Pedro.

Does Moss's diagnosis apply to the weapons example? Here the hypothesis is that the following is not the case: if the USA had thrown its weapons into the sea, the other nuclear powers would have held onto their weapons. It's unclear why just uttering the first sentence in the reverse Sobel sequence should make *this* salient in the relevant way!!!

3. A puzzle for von Fintel

Why are the following discourses so obviously fine?

(8) If all the nuclear powers threw their weapons into the sea tomorrow, there would be peace; but if the USA threw its weapons into the sea tomorrow, the other nuclear powers would not throw their weapons into the sea, and would be war.
(9) If Sophie had gone to the parade and been stuck behind a tall person, she would not have seen Pedro. But if Sophie had gone to the parade, she would not have been stuck behind a tall person, and would have seen Pedro.

Contrast the following, which looks just as terrible as (3b):

(10) Every Chinese philosopher is at a big conference in Beijing; but every philosopher is non-Chinese and went to the Dean's reception in the Silver Center.

Tentative suggestion: maybe the oddity of the original Reverse Sobel Sequences (1b) and (2b) is due to the fact that if you were cooperative, you'd say something like (8) or (9) instead? (But why?)

4. An argument against the strict analysis

Setup: yesterday at noon, I was debating whether to toss three coins, but eventually decided not to. Coin 1 was double-Headed; coin 2 was double-Tailed; coin 3 was normal. (Coins of this sort do have a tiny chance of landing on their edges.)

Natural things to say about this setup:

(11a) It was very likely that coin 1 would land Heads if I tossed it.
(11b) It was very unlikely that coin 2 would land Heads if I tossed it.
(11c) There was about a 50% chance that coin 3 would land Heads if I tossed it.

(11c) looks problematic for the strict conditional account!

If the operative accessibility relation for (11c) is one that worlds where I decide not to toss the coin bear both to worlds where Coin 3 was tossed and landed Heads and to worlds where Coin 3 was tossed and landed tails, then the chance at noon that the proposition that coin 3 would land Heads if I tossed it is way less than 50% (maybe even 0%)!

But what sort of accessibility relation could there be such that among the no-toss worlds, some (about half) are such that all the toss-worlds accessible from them are Heads-worlds, and others aren't?

CEM: arguments for and against

Cian Dorr

10 October 2017

1. For CEM for counterfactuals: from chance

How likely was it that Coin X would land Heads if it was tossed?

2. For CEM in general: from credence

How confident are you that Coin X would have landed Heads if it had been tossed?

How confident are you that Coin X landed Heads if it was tossed?

3. A response from Bennett

> Admittedly, we often find it natural to say things like 'There's only a small chance that if he had entered the lottery he would have won', and 'It's 50% likely that if he had tossed the coin it would have come down heads'. In remarks like these, the speaker means something of the form A > (P(C) = n)—if the antecedent were true, the consequent would have a certain probability; yet the sentence he utters means something of the form P(A>C) = n. According to me, these have different meanings. When we use one to mean the other we employ a usage that is idiomatic but not strictly correct. (p. 251)

Do we really conflate the things Bennett says we conflate? We'd have to conflate (1) (which looks plausibly true) with (2) (which looks clealry false)!

(1) It is pretty likely right now that if Jim drank arsenic tomorrow, he would be dead by the weekend.
(2) If Jim drank arsenic tomorrow, it would be pretty likely right now that he would be dead by the weekend.

4. For CEM: from our carelessness about scope

Williams: to *express our disagreement* with 'If A, it would be that B', we just say 'If A, it wouldn't be that B'.

- By contrast, if we just want to express our disagreement with 'You must not do X' we do *not* feel we have to say 'You must do X'; we just say 'You may do X'.
- Consider negative answers to yes-no questions, e.g. 'Would this coin have landed Heads if it had been tossed? — No.'
  - Claim (is this right?): saying 'no' here is inappropriate unless you have the kind of evidence that would justify saying 'It would have landed Tails if had been

tossed'. 'No, since it's perfectly fair' is weird, but on the CEM-denying view, it looks like it should be great.

Stalnaker's contrast:

(3) a. He doesn't have to appoint any particular woman; he just has to appoint some woman or other.
    b. He won't appoint any particular woman; he just will appoint some woman or other.
    c. He wouldn't have appointed any particular woman; he just would have appointed some woman or other.

(3a) seems fine. (3b) seems weird. Stalnaker's claim: (3c) is as bad as (3b); but if CEM fails in this case, (3c) should be as good as (3a).

- Finds all these sentences to be unduly 'sloppy' as they stand, Bennett suggests fixing them up as follows.

(4) a. There is no woman of whom it is true that he has to appoint her; he just has to appoint some woman or other.
    b. There is no woman of whom it is true that he will appoint her; he just will appoint some woman or other.
    c. There is no woman of whom it is true that he would appoint her; he just would have appointed some woman or other.

- Leave aside for now the *ad hominem* point against Stalnaker (who was led to assert (4c) by his theory of vagueness). Is the fixed-up argument actually any worse than the original?

5. For CEM: from quantified conditionals

Von Fintel and Iatridou (2002): the following seem equivalent

(5) a. Every student will fail if he goofs off [failed if he goofed off/would fail if he goofed off]
    b. No student will pass if he goofs off [passed if he goofed off/would pass if he goofed off]

But assuming these have the logical forms they seem to have, (5b) does not entail (5a) in a logic without CEM. (What if some of the students are such that it's not the case that they will pass if they goof off, and also not the case that they will fail if they goof off?)

6. A possible response from von Fintel: CEM as a mere presupposition

Von Fintel's suggestion: 'If A, B' *presupposes* 'Either if A, B or if A, not-B'.

> Combine this with a general claim about presupposition projection: if 'x is F' presupposes 'x is G', 'Every H is F' and 'No H is F' both presuppose 'Every H is G'.

Result: both (5a) and (5b) presuppose 'Every student either will fail if he goofs off, or will pass if he goofs off'.

So: (5b) is true *and its presupposition is satisfied,* (5a) is true. (And vice versa.)

This proposal also gives a response to some of the arguments about scope. But it doesn't seem to the arguments about chance and credence.

CEM: more arguments

Topics in Mind and Language: Conditionals

12 October 2017

1. Unfinished logic-related business

(i) Given our background logic, CEM entails And-to-if

1.	A	premise
2.	B	premise
3.	$A > \neg B$	assumption
4.	$\neg B$	1,3, MP
5.	$\neg(A > \neg B)$	2–4, reductio
6.	$(A > B) \vee (A > \neg B)$	CEM
7.	$A>B$	5,6, disjunctive syllogism

(ii) Given our background logic, the combination of And-to-if with Antecedent Strengthening entails Or-to-if (i.e. the material conditional analysis)

(here $\top$ represents some arbitrary tautology, e.g. $B \vee \neg B$)

1.	$A \vee B$	premise
2.	$\top$	classical propositional logic
3.	$\top > (A \vee B)$	1, 2, And-to-if
4.	$(\neg A \wedge \top) > (A \vee B)$	3, Antecedent Strengthening
5.	$\neg A > (A \vee B)$	4, substitution of equivalent antecedents
6.	$\neg A > \neg A$	ID
7.	$\neg A > (\neg A \wedge (A \vee B))$	5, 6, CC
8.	$\neg A > B$	7, consequent weakening

(iii)CEM entails the validity of the following principle:

Disjunction Distribution: $A > (B \vee C) \vdash (A > B) \vee (A > C)$

1.	$A > (B \vee C)$	premise
2.	$(A > B) \vee (A > \neg B)$	CEM
3.	$A > \neg B$	assumption
4.	$A > ((B \vee C) \wedge \neg B)$	1, 3, CC
5.	$A > C$	4, consequent weakening
6	$(A > B) \vee (A > C)$	2–5, proof by cases

- Conversely, the validity of Disjunction Distribution entails that of CEM:

1.	$A > A$	ID
2.	$A > (B \vee \neg B)$	1, consequent weakening
3.	$(A > B) \vee (A > \neg B)$	2, disjunction distribution

(iv)Given the closeness analysis, we can get CEM to come out valid by imposing two further constraints on the closeness relation (in addition to transitivity, connectedness, and weak centering):

Antisymmetry If $u \leq_w v$ and $v \leq_w u$, then $u=v$.

The Limit Assumption If S is a nonempty set of worlds and w is any world, then there is some $u \in S$ such that for every $v \in S$, $u \leq_w v$.

- Indeed, the validity of CEM *entails* that Antisymmetry holds, at least in the case where u and v are both accessible from w. For let p_u be any proposition that is true at u and no other possible world, and let p_v be a proposition true at v and no other possible world. Then given the Closeness Analysis, $(p_u \vee p_v) > p_u$ is true at any world w just in case either (i) neither u nor v is accessible from w, or (i) v is accessible from w and u is not, or (iii) both u and v are accessible from w and it's false that $v \leq_w u$, or (iv) $u=v$.
    - Given Disjunction Distribution and ID,$((p_u \vee p_v) > p_u) \vee ((p_u \vee p_v) > p_v)$ is true at every world w, for any worlds u and v. So if u and v are both accessible from w, this means that one of the following conditions obtain: either $u=v$, or $v \leq_w u$ is false, or $u \leq_w v$ is false. In other words: if $u \leq_w v$ and $v \leq_w u$, then $u=v$.
- Something similar is true for the Limit Assumption.

2. The duality argument against CEM (for counterfactuals)

Duality: 'If A, it might be the case that B' is equivalent to 'It is not the case that: if A, it would be the case that not-B'.

- Or to put it another way: 'Not (if A, might B)' is equivalent to 'If A, would not-B'.
- The left-to-right entailment here is pretty uncontroversial; what's controversial is the right-to-left direction, i.e. the claim that 'If A, would not-B" and 'If A, might B' are *inconsistent*.

If we assume both Duality and CEM, we get the crazy result that 'If A, might B' *entails* 'If A, would B':

1.	If A, might B	Premise
2.	Not (If A, would not-B)	Duality
3.	(If A, would B) ∨ (If A, would not-B)	CEM
4.	If A, would B	2, 3, disjunctive syllogism

So either Duality or CEM has to go.

Why believe Duality? Basic datum: conjunctions of the form 'If A, would not-B, and if A, might B', or 'If A, might not-B and if A, would B' sound terrible.

- First worry about this argument: by parity of reasoning it leads to the conclusion that future-directed 'might' conditionals are dual to 'will' conditionals: e.g. 'If I run I might be elected' is inconsistent with 'If I run I will be defeated'.
- Second worry about this argument: 'It might be raining and it isn't raining' also sounds bad. But it looks like a terrible idea to say that this conjunction is *inconsistent*, since this would make the argument from 'It might be raining' to 'It is raining' valid. Are we sure we're not mistaking this other form of badness for the badness of inconsistency?

Challenge: find an alternative account of the meaning of 'If A, might B' that can explain the badness.

Wide scope option: 'If A, might B' is equivalent to 'It might be that (if A, it would be that B)'

Narrow scope option: 'If A, might B' is equivalent to 'If A, it would be that (it might be that B)'.

Lewis's objection (*Counterfactuals* p. 80): 'If I had looked in my pocket, I might have found a penny' is false in the case where I don't have a penny in my pocket, but on these analyses it's true so long as it's consistent with what I know that there is just a penny in my pocket and also consistent with what I know that there is just a dime there.

- This assumes that 'Might P' is equivalent to 'P is consistent with what I know'. (Or maybe 'P is consistent with what I know'.)
- But this may be too simple. Consider: 'I don't know whether this coin might land heads, because I don't know whether it's a double headed coin'. Sometimes, the set of worlds relevant to the truth of a 'might' claim seems to be narrower than the set of all worlds consistent with what we know—in this case, 'might P' perhaps means something more like 'P is consistent with what I know and with the truth about what kind of coin it is'. Maybe in Lewis's case, the operative 'might' is one on which 'might P' is equivalent to 'P is consistent with what I know and with the truth about how many coins are in my pocket'?
    - However if we want to go for a version of the wide scope option using this kind of 'might', we'd like a story about why 'might' gets constrained in this way.

3. An argument for CEM: from quantified conditionals

Higginbotham (****) notes the apparent equivalence of (1a) and (1b), and raises this as a puzzle for semanticists:

(1) a. Every student will fail if he goofs off
    b. No student will pass if he goofs off

We get the same apparent of equivalence when we consider counterfactuals or past-tense indicatives:

(2) a. Every student would have failed if he had goofed off
    b. No student would have passed if he had goofed off

(3) a. Every student failed if he goofed off
    b. No student passed if he goofed off

Let's focus for now on (2a) and (2b). And let's assume they have the logical forms they seem to have, ignoring the option of securing the equivalences by messing with logical form.

For the argument from (2a) to (2b) to be strictly valid, 'x would have failed if he had goofed off, and x would have passed if he had goofed off' would have to be inconsistent. None of the views we have considered says that this is inconsistent, since none of them says that the schema CNC ('Conditional non-contradiction') is valid:

CNC $\neg((A > B) \wedge (A > \neg B))$

(Note that no view on which ID and Weakening the Consequent are both valid can validate CNC, since the instance $(A \wedge \neg A) > (A \wedge \neg A)$ of ID implies both $(A \wedge \neg A) > A$ and $(A \wedge \neg A) > \neg A$ by Weakening the Consequent.) However, we have a resource for explaining the seeming goodness of the inference from (2a) to (2b), namely the *presupposition of non-vacuity*. According to this idea, an utterance of $A > B$ carries the presupposition that some A-worlds are accessible (from the actual world), which entails that corresponding instance of CNC is true.

The direction we need to focus on, then, is the argument from (2b) back to (2a). This is strictly valid just in case 'Not (x would have passed if he had goofed off), and not (x would have failed if he had goofed off)' is inconsistent. Von Fintel and Iatridou (2002) point out that if CEM is valid, this is inconsistent, and (2b) does imply (2a). If CEM *isn't* valid, it is hard to see why (2b) would even seem to imply (2a): why would we ignore the possibility that some of the students are neither such that they would have passed if they had goofed off, nor such that they would have failed if they had goofed off?

4. A possible response: CEM as a mere presupposition

Von Fintel's suggestion: CEM has the same status as CNC. 'If A, B' *presupposes* 'Either if A, B or if A, not-B'.

> Combine this with a general claim about presupposition projection: if 'x is F' presupposes 'x is G', 'Every H is F' and 'No H is F' both presuppose 'Every H is G'.
>
> Result: both (5a) and (5b) presuppose 'Every student either will fail ssarif he goofs off, or will pass if he goofs off'.

So: (5b) is true *and its presupposition is satisfied,* (5a) is true. (And vice versa.)

This proposal also gives a response to some of the arguments about scope. But it doesn't seem to the arguments about chance and credence.

5. Against CEM: from the similarity account of closeness

Even if we are, like Lewis, pretty liberal about what counts as a comparative similarity relation, it's hard to see how any comparative similarity relation could be antisymmetric.

One could retreat from the view that closeness *is* a comparative similarity relation to the view that there is a comparative similarity relation such that whenever u is more similar to w than v is in that sense, u is closer to w than v is (but not vice versa)!

- Note however that if the Limit Assumption fails for the relevant comparative similarity relation, this sufficient condition means that it's also got to fail for closeness, which means we still can't have CEM. And as Lewis points out, the Limit Assumption fails for many comparative similarity relations.
- Also: the kind of reasoning that drives the chance- and credence-based arguments for CEM fits poorly with the comparative similarity vision.
  - Suppose we tossed a fair coin five times yesterday, starting at noon, and it landed Heads on tosses 3 and 5 and Tails on tosses 1, 2, and 4. Q: How likely was it, at noon, that it would land Heads five times, if it landed Heads at least four times?
  - The kind of reasoning we've been relying on suggests the answer '1/6', based on the fact that there are 6 equiprobable ways for it to land Heads at least four times, of which one involves it landing Heads five times. But if being more similar to actuality is sufficient for being closer to actuality, and the number of coin outcomes matters at all, it looks like the answer's going to be 1/64. So long as the coin doesn't *actually* land Heads every time (something whose chance is 1/64), some worlds where it lands Heads exactly 4 times will be more similar to actuality than any world where it lands Heads 5 times.
- Tentative conclusion: if you want to hold on to CEM, you had better forget about the idea that closeness has anything to do with similarity. We know more about similarity than we know about closeness—for example, even knowing how the coin landed, we just don't know whether all-Heads worlds are closer than four-Heads worlds (our credence is 1/6).

6. Against CEM: 'what could make it true'?

7. Background on vagueness and classical logic

We have been assuming classical logic, including the validity of the schema LEM:

LEM $A \vee \neg A$

If this is valid, then so its its substitution instance

(H) Either Harry is bald, or Harry is not bald

Some philosophers think that (H) must be rejected in the case where Harry is borderline-bald. One of their basic arguments is a 'what could make it true?' argument: the thought is that there is nothing in our use of the word 'bald' that could make 'Harry is bald' true or false.

Danger! If you say that it's *not* the case that (either Harry is bald or Harry is not bald), it seems to follow (by a rule that even those who reject some bits of classical logic in this context tend to like) that Harry is not bald and also not not bald. But this is a contradiction (which is again something that most people don't want to be committed to, even in this context).

But many (most?) philosophers thinking about vagueness accept (H).

The *semantic indecision* theory of vagueness: there are many different properties that are 'admissible interpretations' of the word 'bald', some of which are had by Harry and some of which are lacked by Harry. But (H) is *supertrue* — all the propositions that are admissible (uniform) interpretations of (H) are true. So we can assert (H)

- Many philosophers in this 'supervaluationist' tradition (but not Lewis) add: truth for sentences = supertruth. So a disjunction can be true without having a true disjunct.
- Note that this is specific to 'true' when applied to *sentences*. There is no reason I can see to carry it over to 'true' as as a predicate of *propositions*, or to 'it is true that …' — we can say the same thing about 'Either it is true that Harry is bald, or it is true that Harry is not bald' that we said about H itself.

8. Back to CEM

CEM, arbitrariness, and vagueness

Topics in Mind and Language: Conditionals
12 October 2017

1. Recap: arguments we have considered so far for and against CEM

For:

(i) from certain judgments about objective chance (likelihood)
(ii) from certain judgments about rational credence
(iii) from denials and lack of scope distinctions
(iv) [from quantified conditionals]

Against:

(i) from the strict conditional analysis
- which could in turn be argued for on the basis of Reverse Sobel Sequences

(ii) from 'might'-'would' duality

2. Against CEM: from the similarity account of closeness

Even if we are, like Lewis, pretty liberal about what counts as a comparative similarity relation, it's hard to see how any comparative similarity relation could be antisymmetric.

One could retreat from the view that closeness *is* a comparative similarity relation to the view that there is a comparative similarity relation such that whenever u is more similar to w than v is in that sense, u is closer to w than v is (but not vice versa)!

- Note however that if the Limit Assumption fails for the relevant comparative similarity relation, this sufficient condition means that it's also got to fail for closeness, which means we still can't have CEM. And as Lewis points out, the Limit Assumption fails for many comparative similarity relations.
- Also: the kind of reasoning that drives the chance- and credence-based arguments for CEM fits poorly with the comparative similarity vision.
  - Suppose we tossed a fair coin five times yesterday, starting at noon, and it landed Heads on tosses 3 and 5 and Tails on tosses 1, 2, and 4. Q: How likely was it, at noon, that it would land Heads five times, if it landed Heads at least four times?
  - The kind of reasoning we've been relying on suggests the answer '1/6', based on the fact that there are 6 equiprobable ways for it to land Heads at least four times, of which one involves it landing Heads five times. But if being more similar to actuality is sufficient for being closer to actuality, and the number of coin outcomes matters at all, it looks like the answer's going to be 1/64. So long as the coin doesn't *actually* land Heads every time (something whose chance is 1/64),

some worlds where it lands Heads exactly 4 times will be more similar to actuality than any world where it lands Heads 5 times.

- Tentative conclusion: if you want to hold on to CEM, you had better forget about the idea that closeness has anything to do with similarity. We know more about similarity than we know about closeness—for example, even knowing how the coin landed, we just don't know whether all-Heads worlds are closer than four-Heads worlds (our credence is 1/6).

3. Against CEM: the grounding objection

Consider two coins A and B that have never been tossed. Given CEM, we have the following disjunction:

Either A would have landed Heads if A and B had been tossed and exactly one of them had landed Heads, or B would have landed Heads if A and B had been tossed and exactly one of them had landed Heads

But which one is it? What could *make it the case* that it's this one rather than that one that has the relevant property?

- A time-honored concern! In the late 16th century, Molina and Suarez claimed that God had "middle knowledge" of counterfactuals like 'If this free creature were put in this situation, it would make this choice'. Opponents objected that God could not have such knowledge since there were no facts to be known.

4. Does the grounding objection attack LEM as well as CEM?

So far we have been assuming classical logic, including the validity of the schema LEM:

LEM $A \vee \neg A$

If this is valid, then so its its substitution instance

(H) Either Harry is bald, or Harry is not bald

Some philosophers think that (H) must be rejected in the case where Harry is borderline-bald. One of their basic arguments is a kind of grounding objection: what could make it be the case that Harry is bald, or that Harry is not bald? (Or alternatively

To get a better feel for the objection, consider a long row of people all lined up, starting with someone with zero hairs, ending with someone very hirsute, and where each person has just one more hair than his predecessor. Given that person 0 is bald and person 10000000 is not bald, LEM (together with other less controversial rules) leads to the following disjunction: either

Person 0 is bald and person 1 is not bald, or
Person 1 is bald and person 2 is not bald, or …

…or person 999999 is bald and person 1000000 is not bald.

If this is true, then so is the following claim: there is a last bald person in the sequence. Or alternatively: there is a minimum number of hairs compatible with not being bald. But what could make it the case that any particular number has this status?

But there are also powerful considerations in *favour* of LEM. Most obviously, it can be argued for on the basis of the law of non-contradiction, as follows:

1. $\neg(A \wedge \neg A)$	LNC
2. $\neg A \vee \neg\neg A$	1, De Morgan
3. $\neg A \vee A$	2, double negation elimination

So if on reflection we decide we want to keep LEM, there is reason to be suspicious of the grounding objection to CEM.

5. An influential response to the grounding objection to LEM: semantic indecision

Proponents of this view concede something to the grounding objection: although for every property that's entailed by having a billion hairs there is a minimum number of hairs consistent with lacking that property, nothing in our use of the vague word 'bald' singles out any *particular* such property for a unique semantic role. Our use of the word involves **semantic indecision**: there are many different properties all of which count as *admissible interpretations* of the word 'bald'. They have different cutoffs; some of are had by Harry and some are not.

- But many sentences involving the word 'bald' are *supertrue*: they are true on *all* the admissible interpretations. (H) is one example. 'There is a minimum number of hairs consistent with not being bald' is another.
- Supertruth is enough to license assertion once pragmatic factors are cleared away. So we should assert (H), since we should believe *all* the propositions that are its admissible interpretations.
- Some philosophers in this tradition (but not Lewis!) add: truth = supertruth. So a disjunction can be true without having a true disjunct.
  - NB: this is specific to 'true' when applied to *sentences*; it does not carry it over to 'true' as a predicate of *propositions*, or to 'it is true that …'.
  - We can say the same thing about 'Either it is true that Harry is bald, or it is true that Harry is not bald' that we said about H itself.
- This story applies to all kinds of parts of speech, not just adjectives. In particular it can also apply to *names*, like 'baldness'. So we shouldn't, strictly speaking, be asserting things like 'There is no such thing as baldness', or 'there are multiple properties of baldness', etc. — they are superfalse.

6. Does this suffice as a reply to the grounding objection to CEM?

Supervaluationist response: there are many different binary relations among propositions which are admissible interpretations of ‘if’ (or for the counterfactual version of ‘if’?). For each of these relations R, there is a two place relation A and a three place relation $\leq$, obeying the Limit Assumption and Antisymmetry, such that R(p, q) is true at a world w iff either no P-world bears A to w, or the $\leq_w$-minimal p=world bearing A to w is a q-world. So instances of CEM are supertrue.

Worry: does this secure the kind of *ignorance* we want?

CEM, vagueness, and uncertainty

Topics in Mind and Language: Conditionals
12 October 2017

1. Unfinished business: vagueness and ignorance

Consider what happens when we apply the supervaluationist approach to attitude verbs like 'know', 'be pretty certain that', etc.

Answer: typically, when 'Harry is bald' is neither supertrue nor superfalse and you know a reasonable amount about his head, 'You know that Harry is bald' and 'You know that Harry is not bald' will also be neither supertrue nor superfalse. Among the admissible interpretations are propositions you know are true, propositios you know are false, and propositions you don't know either way. Indeed, the set of propositions covers a whole spectrum of degrees of confidence.

So this view predicts that 'I don't know whether Harry is bald' is not a good thing to assert. Similarly for 'I am around X% confident that Harry is bald', and maybe also 'No-one knows who the last bald person in this Sorites sequence is'.

- Is this a good prediction? Reactions vary widely. Some opponents take it to be *obviosly* true that we don't know whether Harry is bald, and proceed from there.
- A not-so-good argument: you can't know whether Harry is bald: if you knew, why wouldn't you just answer 'Yes' or 'No' to 'Is Harry bald?'? Supervaluationist answer: both would be misleading, since they would lead people to believe/disbelieve all (or many) of the admissible interpretations, including both true and false ones.

A tempting generalisation: when you know that a sentence S is neither supertrue nor superfalse, 'You know that S' and 'You know that not-S' are neither supertrue nor superfalse.

- If this were correct, it would be problematic for the CEM defender trying to respond to the "grounding worry". The CEM defender wants to assert things like 'I don't know whether this coin would have landed Heads if tossed' (and even 'I'm about 50% sure that this coin would have landed Heads if tossed').
- However, it's not clear that the supervaluationist approach supports the idea that the tempting generalisation is true in all cases. A simple example: 'Princeton has a population of around 16,000'. (Suppose there are two admissible interpretations of 'Princeton', namely Princeton Borough and "Greater Princeton".).

Are indicative conditionals material conditionals?

Topics in Mind and Language: Conditionals
26 October 2017

1. The material conditional theory of indicative conditionals

An indicative conditional of the form 'If ϕ, ψ' is true at all and only those possible worlds where ψ is true or ϕ is not true.

- We can get this as a limiting case from the variably strict theory by taking the accessibility relation to be one that every world bears to itself and no other world.

2. Reasons for thinking 'If P, Q' entails 'P ⊃ Q'

This follows (given classical logic) from the validity of Modus Ponens for the indicative conditional. $P>Q$ and P together imply Q, so $P > Q$ is inconsistent with $P\wedge\neg Q$, so $P > Q$ implies $\neg(P\wedge\neg Q)$.

3. Arguments for the material conditional theory

The Or-to-if inference: 'P or Q; therefore if not-P then Q'.

- Seems valid! 'It's either a dog or a cat; so if it's not a cat it's a dog'.
- Or-to-if is valid if the indicative conditional is like ⊃ in classical logic in obeying the metarule of *conditional proof*: whenever $P_1,\ldots,P_n \vdash Q$, $P_1,\ldots,P_{n-1} \vdash P_n > Q$

$P\vee Q, \neg P \vdash Q$	(disjunctive syllogism)
$P\vee Q \vdash \neg P > Q$	(conditional proof)

- Gibbard's proof: Or-to-if can be derived from uncontroversial principles of conditional logic together with the *Import-export* principle, according to which 'If A, then if B then C' is equivalent to 'If A and B, then C'. (Actually we only need the right-to-left, 'import' direction of this.).

$\vdash ((P \vee Q) \wedge \neg P) > ((P \vee Q) \wedge \neg P)$	(ID)
$\vdash ((P \vee Q) \wedge \neg P) > Q$	Weakening the consequent
$\vdash (P \vee Q) > (\neg P > Q)$	(Import-Export)
$P \vee Q \vdash \neg P > Q$	(MP)

  A worry about this argument: Import-Export feels valid for counterfactuals too, but Or-to-if seems ridiculous for them!

4. A challenge to the material conditional theory:

Objection: asserting 'If P, Q' in a situation where you just know not-P, or just know Q, seems very bad and misleading. But according to the material conditional theory, you know it's true in this kind of case.

5. An analogy

According to orthodoxy, ‘P or Q’ is entailed by P and entailed by Q. One might object to the orthodoxy on the grounds that when you just know one of P and Q and have no evidence about the other, asserting ‘P or Q’ would be very bad and misleading, even though according to orthodoxy you’re in a position to know that it’s true.

Misleading how? Saying ‘P or Q’ will generally lead hearers to conclude that you don’t know either way.

Response: a sentence can be misleading and bad to assert even if it’s true.

- Strategic point: because of this obvious and pervasive phenomenon, objections of the form ‘This theory should be rejected because it lacks a good explanation of why sentence S is bad to assert’ are hard to make stick. The other phenomenon — something that’s good to assert although false — is much rarer. So it might be easier to object to the material conditional theory on the grounds that it says certain things are false (or likely to be false) that are in fact good to assert.
- However, ‘It is not the case that if P, Q’ is pretty stilted so the fact that materialism makes it so hard for this to be true is not immediately terrible. But maybe we can do better by considering negative answers to conditional questions?

Pragmatic techniques for defending materialism

Topics in Mind and Language: Conditionals
26 October 2017

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- We can get this as a limiting case from the variably strict theory by taking the accessibility relation to be one that every world bears to itself and no other world.

2. Reasons for thinking 'If P, Q' entails 'P ⊃ Q'

This follows (given classical logic for 'or' and 'not') from the validity of Modus Ponens.

1.	$P > Q, P \vdash Q$	MP
2.	$Q \vdash \neg P \vee Q$	$\vee$-intro
3.	$P > Q, P \vdash \neg P \vee Q$	1,2
4.	$P > Q, \neg P \vdash \neg P$	reiteration
5.	$\neg P \vdash \neg P \vee Q$	$\vee$-intro
6.	$P > Q, \neg P \vdash \neg P \vee Q$	4,5
7	$P > Q \vdash \neg P \vee Q$	3,6, $\vee$-elim

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A worry about this argument: Import-Export feels valid for counterfactuals too, but Or-to-if seems ridiculous for them! Hmm.

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Misleading how? Saying 'P or Q' will generally lead hearers to conclude that you don't know either way.

Response: a sentence can be misleading and bad to assert even if it's true.

6. Grice on conversational implicature

Picture: people generally expect one another to be not only literally truthful, but also cooperative. Hearers thus can be expected to make many inferences over and above believing the literal content of the sentence. Speakers can anticipate this and exploit it to convey additional information.

- 'His handwriting is good and he is always on time to class'

Grice offers some "Maxims" which capture what's involved in being cooperative (in the way we expect). They can all be used to generate implicatures: from the fact that someone says this instead of that, we will draw the conclusion that saying that would

> Maxim of Quality. Make your contribution true; so do not convey what you believe false or unjustified.
>
> Maxim of Quantity. Be as informative as required.
>
> Maxim of Relation. Be relevant.
>
> Maxim of Manner. Be perspicuous; so avoid obscurity and ambiguity, and strive for brevity and order.

In the case of 'or', the key maxims are those of quality, quantity and relation. 'P' and 'Q' will almost always be relevant whenever 'P or Q' is, and they are more informative (if true). So from the fact that someone says 'P or Q' rather than 'P' or 'Q' we will generally conclude that they don't believe P and don't believe Q.

- It's a bit complicated. E.g. maybe they actually do believe P but not on the basis of evidence that they think you'd be convinced by, and that's why they are saying only 'P or Q'. Or maybe the obstacle is one of knowledge rather than belief, or lack of high probability — 'Either I'll win the lottery or I won't be able to afford to go on a Safari next year'.
- Thus it's not as simple Jackson's 'Assert the stronger (when probabilities are close)'. [the stronger of what?]

A key feature of converstional implicatures: cancelability.

7. Can conversational implicature save the material conditional theory?

Yes, according to Grice.

Objections from Jackson:

1. It can be OK to assert 'If P, Q' even when one is pretty sure that ¬P.
2. It can be OK to assert 'If P, Q' even when one is pretty sure that Q.
3. Given materialism, 'Either (If P, Q) or (If Q, R)' is a logical truth. Hard to find a pragmatic theory that explains why things of this form are so often bad to assert without having the obviously wrong consequence that logical truths are never good to assert.
4. Sentences that are logically equivalent according to the materialist are assertible under very different conditions in a way that Grice's theory can't explain. Jackson's example: 'Not P and if P, Q' and 'Not P and if P, R'.
5. An argument from 'classroom experience'!?

(1) and (2) don't look problematic given how Grice is really thinking — they are anyway not problematic for the view that whatever explains the assertability facts for disjunctions also explains the assertability facts for conditionals.

(3) I'm not sure about — disjunctions of conditionals with different antecedents are just really hard to process.

(4) is tricky - speeches of the form 'P and if not P then Q' are weird; there's a sense that something is being taken back. But Bennett has a case that really does seem problematic for Grice, namely the contrast between 'If P, not Q' and 'If Q, not P'.

More on materialism

Topics in Mind and Language: Conditionals

31 October 2017

1. Recap

- Materialism
- An argument for materialism: or-to-if seems valid
- An objection to materialism: 'If P, Q' is often *unassertable* when according to materialism it's true (and believed/known to be true!).
- Grice's response: *conversational implicature*
- An objection to Grice's response: 'If P, not Q' can be assertable when 'If Q, not P' isn't.

2. Conventional implicature

Another idea of Grice's: there are special linguistic conventions according to which certain used are used to signal certain things *in a way that does not affect truth conditions*.

> 'She is poor but happy'. The use of 'but' signals that there is some kind of contrast.
>
> 'Even Gödel could prove this'.

Why not instead say that 'P but Q' is *false* if there is no contrast, and that 'Even X could do it' is false if X is particularly good at doing it? One argument concerns embeddings. 'Every task that even John could do will get done' is much stronger than 'Every task that John could do despite being particularly bad at will get done'!

> Conventional implicature 'escapes embeddings'.
>
> Conventional implicature is not (easily) cancellable.
>
> Conventional implicature is a similar sort of phenomenon to presupposition (as discussed earlier) — maybe the very same.

3. Jackson's proposal

'If P, Q' carries the conventional implicature that the speaker has a high level of credence in $P \supset Q$ that is *robust with respect to* P.

Gloss 1: x has a high credence in A that is robust with respect to B if and only if x has high credence in A, and x would continue to have high credence in A if x learnt that B (and nothing more).

Gloss 2: x has high credence A that is robust with respect to B if and only if x's credence in A, and the ratio of x's credence in A-and-B to x's credence in B, are both high.

- This ratio is sometimes called x's "conditional credence" in A given B.

- Note that ($P \supset Q$) and P is logically equivalent to (P and Q). So on Gloss 2 the conventional implicature of 'If P, Q' is equivalent to the speaker having high conditional credence in Q given P.
- A case that pulls the two glosses apart: 'If Reagan is working for the CIA no-one will ever find out' (Lewis). Faced with these kinds of examples, Jackson later went for Gloss 2.

A surprising feature of Jackson's view: according to him, 'P or Q' *conversationally* implicates that the speaker has a high confidence in P or Q that is robust both with respect to P and with respect to Q. So the bar to asserting 'P or Q' is generally *higher* than the bar to asserting 'If not-P, Q'.

4. Confidence-theoretic arguments against materialism

(1) I'm moderately confident that if they went out they went to the movies, but I am very confident that they didn't go out.
(2) It's unlikely that Smith won big if he won
(3) It's starting to look like Smith won big if he won

Not just an idiosyncracy of sentences that contain both 'if' and an attitude verb:

(4) If they went out they went to the movies.
— You're probably right / I'm pretty confident of that / There's a good chance that they did / That's fairly likely to be true / I'm moderately inclined to agree with what you just said…

A similar argument can be made using desire-like (vs. belief-like) attitudes:

(5) I'm hoping I've won a blue car if I've won a car

5. The 'Ramsey Test'

> If two people are arguing 'If A will C?' and are both in doubt as to A, they are adding A hypothetically to their stock of knowledge and arguing on that basis about C. . . . We can say they are fixing their degrees of belief in C given A. (Ramsey 1929: 143)

Conditional degree of belief (a.k.a. conditional credence): $Cr(C \mid A) = Cr(A \wedge C)/Cr(C)$ when $Cr(C) = 0$.

Natural interpretation: your credence in the proposition expressed by 'If A, C' should equal your conditional credence in Q given P, where Q is the proposition expressed by 'C' and P is the proposition expressed by 'A'.

Adam's Thesis: 'degrees of assertability'

Materialism: wrap-up

Cian Dorr

1 November 2017

1. Rebutting the argument from the or-to-if inference

An analogy with the epistemic modal 'must'.

The following arguments have a ring of validity, no?

> It's either raining or snowing
> But it's not raining
> So, it must be snowing

> All men are mortal
> Socrates is a man
> So, Socrates must be mortal

What's going on here…?

> <this bit of the handout will be filled in later>

2. Another family of arguments for materialism: from embeddings

Another strategy for arguing for materialism looks at certain complex sentences that embed indicative conditionals, e.g. conjunctions of indicative conditionals. In some cases, these seem to have the weak truth conditions that materialism would predict.

(1) If the die landed on an odd number it landed on 1 or 3, and if it landed on an even number it landed on 2

- Seems equivalent to 'The die landed on 1 or 2 or 3'. If turns out it landed on 1, we don't say 'OK, so the first conjunct was true, but what about the second conjunct'?[1]

3. A warning

We have already seen ample reason to think that there is lots of context-sensitivity in conditionals. It is a completely live option to think that this is *sometimes* resolved in a way that makes an indicative conditional tantamount to the corresponding material conditional, but not always.

- For example, we might plug the trivial accessibility relation (on which each world only sees itself) into the closeness analysis or the analysis

[1] This example is from Michael McDermott, 'On the Truth-conditions of Certain "If"-sentences'. However McDermott is not himself a Materialist—he favors a kind of three-valued semantics where conditionals with false antecedents are gappy and the conjunction of something gappy with something true is true.

- In the setting of the closeness analysis, ‘If P, Q’ is guaranteed to agree with ‘P $\supset$ Q’ if the operative accessibility relation is one that not-P worlds never bear to P-and-not-Q worlds.

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Probabilities and the Equation

Topics in Language and Mind: Conditionals
November 7, 2017

1. Probability functions

When f is an n-place function and S is a set, S is closed under f iff whenever $x_1 \in S$, …, and $x_n \in S$, and $\langle x_1,\ldots,x_n\rangle$ is in the domain of f, $f(x_1,\ldots,x_n) \in S$.

A set of propositions is **Boolean-closed** (=closed under finitary Boolean operations) iff it is closed under ¬ (the function that maps every proposition to its negation), ∧ (the function that maps any two propositions to their conjunction), and ∨ (the function that maps any two propositions to their disjunction).

A **probability function** is any function Pr from some Boolean-closed set of propositions D to real numbers, such that for any A, B ∈ D

(i) Pr(A) = 1 if A is logically necessary

(ii) Pr(A) ≤ Pr(B) if A logically entails B

(iii) Pr(A∨B) = Pr(A) + Pr(B) if A and B are logically incompatible

Commentary:

- To make sense of the definition we need to understand some notions of "logical necessity", "logical entailment" and "logical incompatibility" that apply to propositions. Any one of these could be defined in terms of the others in obvious ways. It's controversial whether this should just be identified with metaphysical necessity.
- In mathematics, probability functions are standardly identified with functions from *subsets of some given set* to real numbers. If we assumed that propositions are sets of worlds, our definition would be a special case of this; but there is no need to make this assumption if we prefer not to.
- In mathematics, an additional 'countable additivity' condition is often built into the definition.
- Some philosophers instead use "probability function" to denote functions from *the sentences of some language* to real numbers. Given an interpretation of the language which never maps logically inequivalent sentences to the same proposition, you can go back and forth.

Some noteworthy consequences of this definition (for all A, B in the domain of Pr):

(i) If A and B are logically equivalent, Pr(A) = Pr(B)

(ii) Pr(A) ≤ 1

(iii) Pr(A) ≥ 0

(iv) $\Pr(A) + \Pr(\neg A) = 1$
(v) $\Pr(A \wedge B) + \Pr(\neg A \wedge B) = \Pr(B)$
(vi) $\Pr(A \vee B) = \Pr(A) + \Pr(B) - \Pr(A \wedge B)$
(vii) If $\Pr(A \wedge B) = 0$, $\Pr(A) \leq \Pr(\neg B)$
(viii) If $\Pr(A \vee B) = 1$, $\Pr(\neg A) \leq \Pr(B)$

When Pr is a probability function, $\Pr(A \mid B)$ is defined to be $\Pr(A \wedge B) \,/\, \Pr(B)$ whenever $\Pr(B)>0$.

> *Exercise:* show that $\Pr(A) = \Pr(A \wedge B \mid B)\Pr(B) + \Pr(A \wedge \neg \mid B)\Pr(\neg B)$ whenever $0<\Pr(B)<1$

When Pr is any probability function with domain D and B is any proposition for which $\Pr(B)>0$, $\Pr_B$ is defined to be the function on D such that for any $A \in D$, $\Pr_B(A) = \Pr(A \mid B)$.

> *Exercise:* show that $\Pr_B$ is a probability function.

2. Objective chances

Let the objective chance function at *t*, Ch_t, be the function that maps every proposition to its objective chance at *t*.

Claim: for every *t*, Ch^t is a probability function.

Facts entirely about (the intrinsic character of) history before *t* have chance 1 at *t*.

How do chances at different times relate to one another? At least in the special case where the complete truth H about (the intrinsic character of) history between *t* and *t*+ has chance >0 at *t*, we have: $Ch^{t+} = Ch^t_H$. ('Chances evolve by conditionalization').

3. Credences and probabilism

Credence = degrees of belief = degrees of confidence, measured on a scale from 0 to 1.

- The credence function of S at *t* = the function that maps every proposition (in which S has a credence at *t*) to S's credence in it in *t*.

What is it to have a certain credence in a proposition? A hard question for philosophy of mind. But it is worth mentioning an influential "behaviourist" tradition which identifies having a credence with having a certain behavioral dispositions. Cartoon version: to have credence *x* in P is to be willing to pay anything up to $*x* for a bet that pays $1 if P and $0 otherwise.

Do ordinary people actually have credences in most of the propositions they think about? The fact that credences are real numbers makes some philosophers think not. But maybe if you are the kind of philosopher who is comfortable with 'There is a least number of hairs consistent with not being bald', you should be similarly comfortable with 'There is a real number that is Joe Schmoe's current credence that it will rain tomorrow'.

Probabilism: at any time, the credence function of an (ideally) rational person is a probability function.

- Setting aside objections to the claim that rational people should have credence functions at all, the most prominent challenges to probabilism target its implication that if one is ideally rational one assigns credence 1 to all logically necessary propositions.

- However those who dispute this are often happy to treat it as a useful idealization.

(Incremental) Bayesianism: if S has credence function Cr at t and is (ideally) rational throughout the interval from t to t+, then for some proposition E, S's credence function at t+ is Cr_E. ('Credences evolve by conditionalization').

- What proposition should one conditionalize on? Bayesians say: "on one's total evidence". But 'evidence' here is used in a somewhat technical sense.
- Probabilists who reject Incremental Bayesianism want to allow for (i) rational forgetting, and/or (ii) evolution of credence without becoming absolutely certain of anything. 'Bayesian' is often used loosely in a way that covers some alternatives to Incremental Bayesianism.

The Equation and Lewis's triviality results

Topics in Language and Mind: Conditionals

November 9, 2017

1. Unfinished business

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This suggestive remark can be taken in various directions. For example:

Adams's Thesis: the 'degree of assertability' of an indicative conditional (for a speaker at a time) equals the speaker's conditional degree of belief in its consequent given its antecedent.

- What is a "degree of assertability"? I don't know.

3. The Equation

> **Schematic Credence Equation:** Necessarily, at any time, any (ideally) rational person's credence that if P, Q equals their conditional credence in Q given P, provided their credence in P is positive.

4. The Chance Equation

> **Schematic Chance Equation:** Necessarily, at any time, the chance that if it were the case that P it would be the case Q equals the conditional chance of Q given P, provided the chance of P at that time is positive.

- For example: if the proposition that the coin was tossed at 12.01 had a chance of 0.2 at noon, and the proposition that the coin was tossed and landed Heads had a chance of 0.1 at noon, the conditional chance at noon of the latter given the former was 0.5; so according to the Chance Equation, the proposition that the coin would have landed Heads if it had been tossed had a chance of 0.5 at noon.

5. Controlling for context sensitivity

Our formulations of the Equation and the Chance Equation are not ideally clear, given that we have seen some good reasons to take seriously the idea that 'if' sentences are systematically context sensitive.

- On the strong interpretations of the Equations, there is a *single* reading of 'if P, Q' or 'if it were the case that P it would be the case that Q' under which they hold for *every* time (and in the case of the Credence Equation, for every person). But we can also consider a weaker interpretation, where we are allowed to associate different resolutions of the context sensitivity of the conditionals with different times (or different $\langle$person, time$\rangle$ pairs).
- For example, in the case of counterfactuals, we considered a picture where for every time, there is a distinct interpretation of 'If P it would be that Q' where the accessible worlds are those that match history up to that time. Given this semantics, we might expect that the relevant interpretation when we are applying the Chance Equation to t is the one where accessibility is match up to t.
- In the case of indicatives, we considered a picture where for every person and time, there is a distinct interpretation of 'If P, Q' where the accessible worlds are those compatible with that person's knowledge/evidence at that time. Given this semantics we might expect that the relevant interpretation when we are applying the Credence Equation to S at t is the one where accessibility is compatibility with what S knows at t.

To help us avoid running together strong and weak interpretations, let's introduce some definitions. Here, $\Rightarrow$ is any binary operation (function from pairs of propositions to propositions); Pr is any probability function; and **R** is any set of probability functions.

$\Rightarrow$ **is equational for Pr** iff whenever Pr(A)>0 and Pr(B)≥0, Pr(A$\Rightarrow$B) = Pr(B | A).

$\Rightarrow$ **is equational for R** iff $\Rightarrow$ is equational for every Pr $\in$ R.

Now we can distinguish:

> **Invariantist Chance Equation:** Some binary operation $\Rightarrow$ is equational for the set of all probability functions that could be the chance function at some time. The counterfactual 'if' has a reading where it expresses such a $\Rightarrow$.
>
> **Invariantist Credence Equation:** Some binary operation $\Rightarrow$ is equational for the set of all probability functions that could be the credence function of a ideally rational person at some time. The indicative 'if' has a reading where it expresses such a $\Rightarrow$.

And on the other hand:

> **Contextualist Chance Equation:** In a context where t is salient, the counterfactual 'if' is naturally interpreted as expressing a binary operation that is [necessarily?] equational for the chance function at t.
>
> **Contextualist Credence Equation:** In a context where person S and time t are salient (e.g. when S is the speaker and t is the time of speech), the indicative 'if' is naturally interpreted as expressing a binary operation that is [necessarily?] equational for S's credence function at t if S is rational at t.

6. Lewis-style triviality results

Lewisian Lemma: Suppose Pr is a probability function and A and B are propositions such that A entails B and $0<\Pr(A)<\Pr(B)<1$, and $\Rightarrow$ is a binary operation that is equational for Pr. Then $\Rightarrow$ is not also equational for $\Pr_B$.

- *Consequence:* no binary operation is equational for a set of probability functions such that some member of the set is both derived from another by conditionalisation and takes values other than 0 and 1.
- Given that rational credences *can* evolve by conditionalisation (an assumption much weaker than Incremental Bayesianism) this refutes the Invariantist Credence Equation.
- Given that chances evolve by conditionalization on history, this refutes the Invariantist Chance Equation.

Proof of Lewisian Lemma:

1. Given Pr, A, B, and $\Rightarrow$ as described: let C be $A \vee \neg B$.
2. Note that $0<\Pr(C)<1$ and $0<\Pr_B(C)<1$.
3. Since $\Pr(C)>0$ and $\Rightarrow$ is equational for Pr, $\Pr(C \Rightarrow \neg B) = \Pr(\neg B \mid C) = \Pr(\neg B \wedge C)/\Pr(C)$ $= \Pr(\neg B)/\Pr(C) > \Pr(\neg B)$.
4. Hence $\Pr(B \wedge (C \Rightarrow \neg B)) > 0$.
5. So, $\Pr_B(C \Rightarrow \neg B) = \Pr(B \wedge (C \Rightarrow \neg B))/\Pr(B) > 0$.
6. But $\Pr_B(\neg B \wedge C) = \Pr(\neg B \wedge C \wedge B)/\Pr(B) = 0$.
7. Hence $\Pr_B(\neg B \mid C) = \Pr_B(\neg B \wedge C)/\Pr_B(C) = 0$.
8. So by lines 4 and 6, $\Rightarrow$ is not equational for $\Pr_B$.

7. Lewis's first three theorems

They are much weaker (and thus less telling against the Invariant Equations) than our Lemma, but here goes:

First triviality theorem: no binary operation $\Rightarrow$ is equational for the class of all probability functions on any Boolean-closed domain D that contains three logically possible, logically incompatible propositions.

- This follows from the Lemma, since given such a D and appropriate propositions A, B,C, we can construct Pr such that 0<Pr(A)<Pr(A∨B)<1.
- But this result is quite boring and obvious, since it is easy to show that for any such D, there is a probability function Pr whose range is {0, 1/3, 2/3, 1} and which assigns 1/3 to two logically incompatible propositions. But no ⇒ can be equational for such a Pr, since if Pr(A) = Pr(B) = 1/3 and A and B are logically incompatible, Pr(A | A∨B) = 1/2, so Pr(A⇒B) would have to be 1/2 if any operation ⇒ were equational for Pr.

Second triviality theorem: no binary operation ⇒ is equational for any class of probability functions on the same domain that is closed under conditionalisation and contains some nontrivial probability function (one such that 0<Pr(A∧B)<Pr(B)<1 for some A, B).

- This follows from the Lemma is much weaker than it and less significant. Not so plausible that rational credence functions/ possible chance functions are closed under conditionalisation on just any old proposition!

Third triviality theorem: no binary operation ⇒ is equational for any class of probability functions on the same domain that is closed under conditionalisation on the members of some finite partition.

8. What about giving up on conditionalisation?

Say that a probability function is *regular* iff whenever P(A)=1, A is logically necessary. Some hold that all rational credence functions are regular (on 'anti-Cartesian' grounds). This provides independent reason to doubt that any rational credence function comes from any other by conditionalisation.

Also: in the real world it seems that the complete truth about any interval of history is always so fully specific that it can't have any chance higher than 0. This provides indepedent reason to doubt that the chance function at any time comes from the chance function at any other time by conditionalisation.

However there are strengthenings of Lewis's results that suggest that the problem isn't just with the particular operation of conditionalisation. For example:

Hall's triviality result: if ⇒ is equational for two distinct, non-trivial probability functions Pr and Pr′, then Pr and Pr′ are "near-orthogonal", in the following sense: for every ε>0, there is a proposition A such that Pr(A)<ε and Pr′(ε)>1–ε.

It's pretty hard to believe that rational changes of mind always involve moving from one credence functions to another that is near-orthogonal to it!

Stalnaker's static triviality result

Cian Dorr

14 November 2017

1. The concept of probabilistic independence

Definition: When Pr is a probability function and A and B are propositions in its domain, A and B are *independent in* Pr iff $\Pr(A \wedge B) = \Pr(A)\Pr(B)$.

Some facts about probabilistic independence:

Fact 1: if either Pr(A) or Pr(B) is 0 or 1, A and B are automatically independent in Pr.

Fact 2: The following are all equivalent:

a) A and B are independent in Pr
b) A and ¬B are independent in Pr
c) ¬A and B are independent in Pr
d) ¬A and ¬B are independent in Pr

- Proof for (b): $\Pr(A\wedge\neg B) = \Pr(A) - \Pr(A\wedge B) = \Pr(A) - \Pr(A)\Pr(B) = \Pr(A)(1 - \Pr(B)) = \Pr(A)\Pr(\neg B)$. Proofs for others are similar.

Fact 3: When $\Pr(B)>0$, A and B are independent in Pr iff $\Pr(A)=\Pr(A \mid B)$.

- So, if $\Pr(Y \mid X) = \Pr(Y)$ and $\Pr(X)<1$ then $\Pr(Y \mid \neg X) = \Pr(Y)$.

Fact 4: if X and Y are independent in Pr and X logically entails Y, then either $\Pr(X) = 0$ or $\Pr(Y) = \Pr(Y \mid X) = 1$.

2. Equationality and independence

Where ⇒ is any binary operation on propositions:

⇒ *obeys MP* iff for any propositions A and B , A⇒B entails A⊃B.

⇒ *obeys And-to-if* iff for any propositions A and B, A∧B entails A⇒B.

Fact 5: If ⇒ obeys MP and And-to-if, then for any A and B, A∧(A⇒B) is logically equivalent to A∧B.

Fact 6: If ⇒ obeys MP and And-to-if, then for any A and B, $\Pr(A\Rightarrow B \mid A) = \Pr(B \mid A)$.

Fact 7: If ⇒ obeys MP and And-to-if and is equational for Pr then for any A and B with $\Pr(A)>0$, $\Pr(A\Rightarrow B \mid A) = \Pr(A\Rightarrow B)$—in other words, A⇒B is independent of A in Pr.

- (Recall from the last handout the following definition: ⇒ is equational for Pr iff for any A and B in the domain of Pr with $\Pr(A)>0$, $\Pr(A\Rightarrow B) = \Pr(B \mid A)$.)

Fact 8: If ⇒ obeys MP and And-to-if and is equational for Pr, then for any A and B such that A⇒B entails A, either $\Pr(A\Rightarrow B) = 0$ or $\Pr(A) = 1$. (From facts 4 and 7.)

- Strategy: we're going to find schemas A* and B* that we can substitute for A and B such that every instance of $A^*\Rightarrow B^*$ logically entails A*, but so long as Pr is nontrivial there are values of A* and B* such that neither $Pr(B^* \mid A^*) = 0$ nor $Pr(A^*)=1$.

3. "Very Limited Antecedent Strengthening"

Recall the following principle (valid in Lewis's conditional logic, and hence in all the other logics we have looked at):

RCV: $A > B, A > C \vdash (A\wedge B) > C$

or equivalently (given CC and weakening of the consequent):

$A > B\wedge C \vdash\!\dashv A\wedge B > C$

or equivalently (given ID):

$A > (A\wedge B\wedge C) \vdash (A\wedge B) > (A\wedge B\wedge C)$

or equivalently as a metarule:

If Strong $\vdash$ Medium and Medium $\vdash$ Weak, then Weak > Strong $\vdash$ Medium > Strong.

4. A conditional that logically entails its own antecedent

Here's our "$A^*\Rightarrow B^*$":

(S) $(A \vee A\Rightarrow B) \Rightarrow A\wedge B$

by RCV, (S) entails $(A \wedge (A \vee A\Rightarrow B)) \Rightarrow A\wedge B$

by substitution in the antecedent, this entails $A \Rightarrow A \wedge B$

by weakening of the consequent, this entails $A \Rightarrow B$,

by disjunction introduction, this entails $A \vee (A\Rightarrow B)$. QED!

5. Upshot

If $\Rightarrow$ is equational in Pr and obeys minimal conditional logic, then for any A and B, either

(a) $Pr(A \vee A\Rightarrow B) = 1$, or

(b) $Pr((A \vee A\Rightarrow B) \Rightarrow (A\wedge B)) = 0$.

If (a), then either (a.1) $Pr(A)=1$, or (a.2) $Pr(A\Rightarrow B \mid \neg A) = 1$

If (a.2) then $Pr(A\Rightarrow B)=1$, so $Pr(B \mid A)=1$, so $Pr(A\wedge B) = Pr(A)$.

If (b), then either (b1) $Pr(A\vee A\Rightarrow B) = 0$ or (b2) $Pr(A\wedge B \mid A\vee A\Rightarrow B) = 0$

If (b1) then $Pr(A) = 0$

If (b2) then $Pr(A\wedge B \wedge (A\vee A\Rightarrow B)) = 0$, so $Pr(A\wedge B) = 0$.

So: for any A and B, either Pr(A) = 0, or Pr(A) = 1, or Pr(A∧B) = 0, or Pr(A∧B)=Pr(A)

So: there are no A and B such that 0<Pr(A∧B)<Pr(A)<1.

- That is: Pr is "trivial".

6. Some possible reactions to this result

(i) Give up altogether on the Equation

(ii) Give up on the idea that conditionals express propositions (up next)

(iii)Weaken the logic by rejecting RCV

- This avenue is explored in Andrew Bacon, 'Stalnaker's Thesis in Context'

(iv)Restrict the Equation

- The important inspiration here is from van Fraassen, 'Probabilities and Conditionals'.

7. More on van Fraassen's result

Required concepts: some propositions are *categorical* ('unconditional', 'just about how things actually are'…); the rest we can call *hypothetical*.

- When A and B are categorical, ¬A, A∧B, and A∨B are categorical too, but the propositions eligible to be expressed by 'If A, B' are not (typically).
- Simple ordinary sentences not involving 'if' express categorical propositions.

Van Fraassen shows that the following are consistent:

> **Restricted Contextualist Chance Equation:** In a context where *t* is salient, the counterfactual 'if' is naturally interpreted as expressing a binary operation ⇒ such that for all categorical A having positive chance at *t* and for all B, the chance at *t* of A⇒B equals the conditional chance at *t* of B given A.

> **Restricted Contextualist Credence Equation:** In a context where person *S* and time *t* are salient (e.g. when S is the speaker and *t* is the time of speech), the indicative 'if' is naturally interpreted as expressing a binary operation ⇒ such that for all categorical A to which S assigns positive credence at t and for all B, the S's credence at *t* in A⇒B equals S's conditional credence at *t* in B given A, provided S is rational.

Non-factualism

Cian Dorr

16 November 2017

1. The view

Nonfactualists about some class of declarative sentences in a certain language say that although sentences in that class are meaningful, they do not express propositions.

- It's already clear that there are meaningful bits of language that don't express propositions—subsentential expressions; questions; imperatives (arguably).
- Some important candidates for this status: moral sentences (those of the form 'X is right/wrong/good/bad'…); aesthetic sentences; 'funny', 'tasty'; epistemological sentences ('this is good evidence for that/reason to believe that'); epistemic modals ('might P', 'must P', 'probably P', 'it's likely that P'); conditionals.

A not-so-great follow-up question: 'What do they express, then, if not propositions?'

Before plunging into that sort of question, we want to consider what it *is* for a sentence (as used by a certain linguistic community) to express a certain proposition.

- Once we have a sense for how we might answer that question we might be in a position to describe some *other* relation between a sentence and a linguistic community that does *not* suffice for the sentence to express any proposition for that community, but is still similar in some ways to the relations that do so suffice so that we can see why it's should count as a way of being meaningful.
- Whether we want to characterise this other relation as a matter of "expressing" something other than a proposition is neither here nor there—"express" here is a term of art.

2. Crash course in intention-based metasemantics

There are many many kinds of answers to the question 'What is it for a sentence to express a proposition for a community?' that suggest all kinds of different generalisations. I'm going to just look at one small family, influenced by David Lewis (*Convention*, 'Languages and Language') and Bennett (*Linguistic Behavior*).

- One of its hallmarks is that it analyses this notion in *mental* terms—expressing P in a community is a matter of the knowledge, beliefs, desires, intentions, … of people in the community.
- A regularity is a *convention* in a community iff (roughly) people in the community generally abide by it; expect each other to abide by it; prefer abiding by it themselves assuming others do; and know all of the above.

First approximation: S expresses P in C iff there is a convention in C that one doesn't utter S unless <u>one believes P</u>.

- Alternative completions:
    - <u>one knows P</u>
    - <u>one intends one's audience to believe that P on the basis of one's saying it</u>
    - <u>one intends one's audience to believe that one believes P</u>
    - <u>P is true</u>
    - <u>one intends that everyone in the conversation should believe P, believe that everyone else believes P; believe that everyone else believes *that*; etc.</u>

3. Generalisations

Each of these views suggests certain generalisations—binary relations between sentences and communities similar in structure to the relations of expressing P, but different.

EG: Suppose we think that the central convention concerning 'Snow is white' is that one does not utter it unless one <u>intends one's audience to believe that snow is white</u>. One could then think, similarly, that one not utter 'Lying is wrong' unless one <u>intends one's audience to disapprove of lying</u>.

- By contrast, the convention for 'I disapprove of lying' will be that one not utter it unless one <u>intends one's audeience to believe that one disapproves of lying</u>. This is totally different!

4. Expressivism

Expressivists about sentences in a certain class hold that the functioning of sentences in that class in the relevant community can be summed up by associating each such sentence S with a certain mental state M(S) which is not the state of believing some proposition.

The idea is that *whatever* the important relation is an ordinary factual sentence expressing P in a community bears to the mental state of *believing P*, that same relation holds between S and M(S).

- An attempt to avoid having to commit to any particular metasemantics.
    - E.g. maybe M('lying is wrong') = disapproval of lying
    - E.g. maybe M('it is probably raining') = having credence at least 0.5 that it is raining
    - E.g. maybe M('if it is raining the ground is wet') = having high conditional credence that the ground is wet given that it is raining.

    - note that this would be particularly natural

5. Truth and falsity for sentences

If being a true/false sentence is just expressing a true/false proposition, then clearly nonfactualists about a certain class of sentences must think that sentences in that class are never true or false.

- Bennett, Adams don't want to do this — they want to say that 'If P, Q' is false when P is true and Q is false.
- My view: 'true sentence' is pretty much a term of art—in natural language 'true' is normally used in constructions like 'She said something true'. So it doesn't matter much what we say.

6. The "Frege-Geach problem" (or challenge)

The challenge: it's not enough to just talk about some family of simple sentences involving X-type vocabulary. We need a theory that gives us a systematic way of characterising the meanings of *all* well-formed sentences involving X-type vocabulary.

- This includes, e.g., negations and disjunctions, including 'mixed' ones. 'Lying is not wrong'; 'Either lying is wrong or gambling is wrong'; 'Either lying is wrong or snow is white'; 'Lying is wrong and gambling is wrong'.
- Also sentences with quantifiers and modals
- And mental state ascriptions. 'She believes that lying is wrong'; 'She is pretty confident that lying is wrong'; 'She hopes that lying is wrong'; 'She is pretty confident that either lying is wrong or gambling is wrong'…

Much of the history of discussion on this point consists in people saying very programmatic things — 'reasons for optimism' rather than theories.

- In the case of conditionals, there has considerable emphasis on the fact that *some* embeddings of conditionals are hard to interpret. OK, but many others are not and we need a theory for them

Developing non-factualism about conditionals

Cian Dorr

20 November 2017

1. Recap: non-factualism in general; expressivism in particular

2. The "Frege-Geach problem" (or challenge)

The challenge: it's not enough to just talk about some family of simple sentences involving X-type vocabulary. We need a theory that gives us a systematic way of characterising the meanings of *all* well-formed sentences involving X-type vocabulary.

- This includes, e.g., negations and disjunctions, including 'mixed' ones. 'Lying is not wrong'; 'Either lying is wrong or gambling is wrong'; 'Either lying is wrong or snow is white'; 'Lying is wrong and gambling is wrong'.
- Also sentences with quantifiers and modals
- And mental state ascriptions. 'She believes that lying is wrong'; 'She is pretty confident that lying is wrong'; 'She hopes that lying is wrong'; 'She is pretty confident that either lying is wrong or gambling is wrong'…

Much of the history of discussion on this point consists in people saying very programmatic things — 'reasons for optimism' rather than theories.

- In the case of conditionals, there has considerable emphasis on the fact that *some* embeddings of conditionals are hard to interpret. OK, but many others are not and we need a theory for them

3. An obvious but problematic strategy (for an expressivist)

Associate operators ('not', 'and', 'or', 'necessarily', 'Joe believes that…') with *functions from mental states to mental states*. When $M(S_1) = m_1$ and $M(S_2)=m_2$, $M(\text{'}S_1 \text{ or } S_2\text{'}) = f_\vee(m_1,m_2)$.

This looks super hard, even for factual sentences!

- What function do you apply to *believing snow is white* to get *believing that snow is not white*? (Note that this is not the same as *not believing that snow is white*, so the function isn't just the ordinary *negation* operation on properties.)
- What function do you apply to *believing snow is white* and *believing grass is green* to get back *believing that snow is white or grass is green*? (Note that this is not the same as *believing that snow is white or believing that grass is green*, so the function isn't just the ordinary disjunction operation on properties.

4. A more flexible, two-part strategy

Part 1: an assignment $[\![\cdot]\!]$ of *compositional semantic values.* $[\![S_1 \text{ or } S_2]\!] = [\![\text{or}]\!]([\![S_1]\!], [\![S_2]\!])$, etc.

Part 2: a *bridging principle* mapping these compositional semantic values onto mental states (if you're an expressivist), or more generally and ultimately, onto whatever the relations between sentences and linguistic communities the nonfactualist proposes to substitute for relations _ *expresses proposition P in* _.

- We can take the compositional semantic values to be whatever it's most convenient to take them to be, so long as we can formulate good bridging principles mapping these entities back to mental states (or whatever).
- The mapping doesn't have to be one-one. Perhaps there are two sentences that, when asserted on their own, would express the exact same mental state, but such that substituting one for another in some more complex sentence would change which mental state it expresses.

5. An example: trivalent semantics for expressivists

Semantic values of sentences are functions from the set of all possible worlds to the three-membered set {0, ½, 1}.

- Semantic values of names are objects; semantic values of predicates are functions from name semantic values to sentence semantic values; semantic values of operators are functions from sentence semantic values to sentence semantic values; etc.
- When S expresses a proposition P, (i.e. a factual sentence) $[\![S]\!]$ the "bivalent" function f such that $f(w)=1$ when the proposition is true at w and $f(w)=0$ otherwise. We assume that $[\![\text{and}]\!]$, $[\![\text{or}]\!]$, and $[\![\text{not}]\!]$ behave in the standard ways when given bivalent functions as arguments.

Bridging principle: a sentence with semantic value f expresses the mental state *having high credence in* $f^{-1}(\{1\})$ conditional on $f^{-1}(\{0, 1\})$.

- For example, suppose we have a sentence S whose semantic value is the function f where $f(w)=0$ if it is raining at w and the ground is not wet, $f(w)=½$ if it is not raining at w, and $f(w)=1$ if it is raining at w and the ground is wet. Then S expresses the mental state *having high conditional credence that the ground is wet given that it is raining*.

Some compositional rules:

$$[\![\text{If}]\!](f, g)(w) = \begin{cases} ½ \text{ if } f(w) = 0 \text{ or } f(w) = ½ \text{ or } g(w) = ½ \\ 0 \text{ if } f(w) = 1 \text{ and } g(w) = 0 \\ 1 \text{ if } f(w) = 1 \text{ and } g(w) = 1 \end{cases}$$

⟦Is at least n% confident that⟧(f)(x)(w)
= 1 if at w x's credence in $f^{-1}(\{1\})$ given $f^{-1}(\{0, 1\})$ is at least n%
= 0 otherwise

- *Consequence:* when A and B are sentences that express propositions P and Q, 'Mary is at least n% confident that if A, B' expresses a proposition that is true exactly those worlds where Mary's credence in Q given P is at least n%.
- Another interesting consequence: When A, B, C express propositions, ⟦If A, if B, C⟧ = ⟦If (A and B), C⟧ (here using the assumption that 'and' behaves standardly on bivalent arguments).

How should ⟦and⟧, ⟦or⟧, ⟦not⟧ work for non-bivalent arguments?

- For ⟦not⟧ there is only one reasonable choice, namely ⟦not⟧(f)(w) = 1–f(w). Otherwise we'd have a difference between ⟦A⟧ and ⟦not not A⟧.
- But for ⟦and⟧ and ⟦or⟧ there are several choices, the most natural of which are the Strong Kleene, Weak Kleene, and McDermott truth tables.

$\wedge_{wk}$	1	½	0
1	1	½	0
½	½	½	½
0	0	½	0

$\wedge_{sk}$	1	½	0
1	1	½	0
½	½	½	0
0	0	0	0

$\wedge_{McD}$	1	½	0
1	1	1	0
½	1	½	0
0	0	0	0

$\vee_{wk}$	1	½	0
1	1	½	1
½	½	½	½
0	1	½	0

$\vee_{sk}$	1	½	0
1	1	1	1
½	1	½	½
0	1	½	0

$\wedge_{McD}$	1	½	0
1	1	1	1
½	1	½	0
0	1	0	0

- Focusing on ⟦and⟧, we have the following rather surprising consequences, for factual sentences A, B, C, D:
  - If ⟦and⟧ is $\wedge_{wk}$, ⟦(If A, B) and C⟧ = ⟦If A, (B and C)⟧
    - In that case 'I am more confident that (if A, B) and C than that C' can be true (even if I'm rational)!
  - If ⟦and⟧ is $\wedge_{sk}$, ⟦(If A, B) and C⟧ = ⟦If (A or not C), (A and B and C)⟧
    - Not so clear if this is bad. But looks worse when we turn to conjunctions of two conditionals.

- If ⟦and⟧ is $\wedge_{McD}$, ⟦(If A, B) and C⟧ = ⟦(not (A and not B)) and C⟧
  - In that case 'I am more confident that (if A, B) and C than that if A, B' can be true!
- Turning to conjunctions of two conditionals, we have the following (exercise!):
  - If ⟦and⟧ is $\wedge_{wk}$, ⟦(If A, B) and (If C, D)⟧ = ⟦If (A and C), (B and D)⟧
  - If ⟦and⟧ is $\wedge_{sk}$, ⟦(If A, B) and (If C, D)⟧ = ⟦If $A \wedge C$ or $A \wedge \neg B$ or $C \wedge \neg D$, (A and B and C and D)⟧.
  - If ⟦and⟧ is $\wedge_{McD}$, ⟦(If A, B) and (If C, D)⟧ = ⟦If (A or C), (B and D) or (A and D) or (B and C)⟧.

To draw out the surprising consequence of the $\wedge_{sk}$ treatment, suppose we have a 100 ticket lottery. Consider the sentence $S_1 \wedge S_2$ = 'If a number under 51 is drawn a number under 50 will be drawn, and if a number over 49 is drawn a number over 50 will be drawn'. Its value is ½ (= $1 \wedge_{sk}$½) at worlds where a number under 50 is drawn; 0 (=$0 \wedge_{sk} 0$) if 50 is drawn; and again ½ (= ½ $\wedge_{sk}$ 1) if the number drawn is great then 50. So it's equivalent to 'If 50 is drawn, 50 won't be drawn'. If you're rational, 'You are 0% confident that $S_1 \wedge S_2$' is true, despite the fact that (assuming you know the lottery is fair), 'You are 98% confident that S_1' and 'You are 98% confident that S_2' are both true. Curious!

Kratzer on 'if'-clauses as restrictors

Topics in Language and mind
28 November 2017

1. True embedding versus merely orthographic embedding

Suppose we are theorising about the sentence

(1) Antony loves Cleopatra

trying to figure out what proposition it expresses, or what compositional semantic value to assign it. An important source of evidence concerns sentences which *embed* (1), such as

(2) a. It is necessary that Antony loves Cleopatra.
 b. Either Antony loves Cleopatra or Cleopatra loves Antony.
 c. Caesar thinks that Antony loves Cleopatra.

We should expect that our compositional semantic value for (1) should fit together with compositional semantics for the other words in these sentence to yield correct semantic values for (2a–c). But our inquiry will go completely off the rails if we expect the same thing in the case of (3a–b):

(3) a. Everyone who loves Antony loves Cleopatra.
 b. Antony loves Cleopatra more than anyone else.

'Antony loves Cleopatra' is not a *constituent* in (3a) and (3b). For example the constituency structure of (3a) seems to be [Everyone [who [loves Antony]]] [loves Cleopatra]; to work out the semantic value of (3a) we only need to know the semantic values of the nodes in this tree.

Similarly, (4) is not a constituent in (5a–b):

(4) Dogs bark.
(5) a. Some dogs bark.
 b. Most dogs bark.

The tree structure of (5b) is [Most dogs] bark, not Most [dogs bark].

Consider finally:

(6) Whenever I go outside, dogs bark

Although it is defensible to think that 'dogs bark' is a constituent in (6), it still seems implausible to think that its meaning can be put together by applying a 'Whenever I go outside' operator to the proposition expressed by (4) in isolation, which is roughly that barking is a characteristic behaviour of dogs. The idea of "generic" or "characteristic" behaviour doesn't seem to enter into the meaning of (6) at all. A possible diagnosis:

when uttered in isolation, (4) includes an unpronounced element GEN contributing this element of meaning: its logical form is something like '[GEN dogs] bark' or maybe 'GEN [dogs bark]'. But in (6), GEN does not appear at all.

2. Conditionals with adverbs of quantification

Consider:

(7) If a man buys a horse, he pays cash for it.
(8) a. Sometimes, if a man buys a horse, he pays cash for it.
b. Always, if a man buys a horse, he pays cash for it.
c. Usually, if a man buys a horse, he pays cash for it.

Weak claim: the complete structure that would be contributed by (7) in isolation is not a constituent in (8a–c).

Strong claim: the string 'if a man buys a horse, he pays cash for it' does not correspond to any constituent in (8a–c). Their real compositional structure is one where 'sometimes' combines with one of the two clauses before they combine with each other—either [Sometimes [if a man buys a horse]] [he pays cash for it], or else maybe [If a man buys a horse] [sometimes [he pays cash for it]].

Evidence for weak claim: the proposition expressed by (7) has a 'generic' flavour which seems absent from, say, (8a).

Evidence for strong claim: the competing tree structures do a better job of predicting the intuitive meanings of (8a)–(8c).

- Strategy: take the 'if' clause and main clause to denote properties of events, and take adverbial quantifiers to denote functions from properties of events to functions from properties of events to propositions. For example: $[\![\text{sometimes}]\!](F)(G)$ = the proposition that some events have both F and G; $[\![\text{usually}]\!](F)(G)$ = the proposition that most events having F have G; etc.
- Given this semantic value, the AdvQ needs to combine semantically with the 'if'-clause and then the result needs to combine with the main clause.

3. Conditionals with modals

Kratzer favours a parallel account of sentences in which 'if'-clauses interact with modal verbs like like 'must', 'can', 'should', 'would', 'have to', 'need to', 'be able to'; modal adverbs like 'probably', 'possibly', 'necessarily'; and modal adjectives like 'probable', 'likely', 'possible', 'necessary'.

Her favoured syntax seems to assign all of (9a–c) the logical form (9d):

(9) a. Probably, if Yog had white, Zog won
b. If Yog had white, Zog probably won

c. Probably, Zog won if Yog had white.
d. [Probably [if Yog had white]] [Zog won]

Semantics for 'probably': in context, 'probably' denotes a function from propositions to functions from propositions to propositions—⟦probably⟧(P)(Q) is true in those worlds where the relevant people's evidence supports a high conditional credence for Q given P. (Or something like that.)

If this is right, then the truth conditions of (9a–c) don't tell us anything about the proposition expressed by the bare conditioal 'If Yog had white, Zog won'.

4. Extending to attitude operators

In this paper Kratzer does not discuss sentences like (10a–b):

(10) a. I am pretty confident that if Yog had white, Zog won.
b. I am pretty confident that Zog won if Yog had white.

Up to now, we have been sentences like these as quite important data points for figuring out what's expressed by 'if Yog had white, Zog won': we argued that the proposition in question must be one that it would make sense to be pretty confident in exactly when one's conditional credence in Zog having won conditional on Yog having white was high. But this is all wrong if we adopt a view where the string 'if Yog had white, Zog won' doesn't correspond to any constituent in (10).

5. Counterfactuals and their chances

For Kratzer the 'would' in counterfactuals behaves like any other modal: (11a) has an underlying structure like (11b).

(11) a. If the coin were tossed, it would land heads
b. [Would [if the coin were tossed]] [it land heads]

Since the 'would' is already there playing the role of the modal that combines directly with the 'if'-clause, (12a) *does* have a constituent corresponding to (11a):

(12) a. It is 50% likely that if the coin were tossed, it would land heads.

So, the Kratzer-style logical syntax does *not* disrupt the arguments we have been making earlier in the course where we used sentences like (12a) to draw conclusions about the propositions expressed by sentences like (11a).

6. Apparently bare condiitonals

When there isn't an overt (audible, visible) modal (or AdvQ) in a sentence with an 'if' clause in it, Kratzer posits that there is a covert (unpronounced) one. Sometimes this is an AdvQ along the lines of 'always' or 'typically' or 'generically', as in (7) (see above). But for standard indicative conditionals, Kratzer proposes that the hidden modal is an *epistemic necessity* operator with a meaning similar to that of the overt modal 'must'. For example, (13a) has the logical form (13b):

(13) a. If the coin was tossed, it landed heads
b. [MUST [if the coin was tossed]] [it landed heads]

'MUST' is context-sensitive like 'must': in our terms, it expresses a function that maps any proposition P to the function that maps any proposition Q to a proposition true in all worlds where the salient people's evidence entails P⊃Q.

- Note that the rest of Kratzer's story would work just as well no matter what she said about the semantics of this silent modal—she could even take it to be the material conditional! The choice of a 'must' like meaning is not fully motivated in the paper we read.
- Kratzer's view allows her to say even when an overt modal is present, there are some cases where the covert modal is present too and is the one that interacts directly with the 'if' clause. So for example (14a) could perhaps have a LF like (14c) as welll as one like (14b):

(14) a. It's possible that Jane was there if Mary was there
b. [Possible [if Mary was there]] [Jane was there[]
c. [Possible ∅] [[MUST [if Mary was there]] [Jane was there]]

These would be predicted to have quite different meanings—(14b) will be equivalent to 'it's possible that Jane and Mary were both there', while (14c) will be equivalent to 'It's possible that Mary couldn't have been there without Jane' (on an epistemic reading of the 'couldn't').

7. Stacking 'if' clauses

Challenges to Modus Ponens

Cian Dorr

30 November 2017

1. Recap

2. Apparent counterexamples to Modus Ponens

A counterexamples to Modus Ponens would be a case where we have three sentences A, B, C, such that A and B are true, C is not true, and A is $\ulcorner$If B, C$\urcorner$ (or $\ulcorner$If B, then C$\urcorner$).

- Given a Kratzer-style syntax a question immediately arises about what the final condition means, given that we want to think of sentences not as mere strings of letters but as structured entities. But let's barge ahead for now and see how this plays out.

Orthodoxy says there are no counterexamples to Modus Ponens. But here are some seeming candidates.

(i) The paradox of gentle murder (Forrester): A:'If John kills his mother, he ought to kill her gently'. B:'John will kill his mother'. C:'John ought to kill his mother gently'.

(ii) (A slight variant:) A 'If we're going to leave tomorrow we should start packing now'; B: 'We're going to leave tomorrow'; C: 'We should start packing now'.

(iii)(McFarlane and Kolodny): A: 'If the miners are in shaft 1 we should flood shaft 2'; B: 'The miners are in shaft 1'; C: 'We should flood shaft 2'.

(iv)(Yalcin): A: 'If the marble is big, it's likely red'; B: 'The marble is big'; C: 'The marble is likely red'.

(v) (A slight variant): A: 'If it's Tuesday we must be in Paris'; B: 'It's Tuesday'; C: 'We must be in Paris'.

(vi)(A different kind of variant): A: 'If the marble is big, I'm confident that it's red'; B: 'The marble is big'; C: 'I'm confident that the marble is red'. [Also consider versions with 'know' and 'think'.]

(vii)(McGee): A: 'If a Republican wins, then if Reagan doesn't win, Anderson will win'; B: 'A Republican will win'; C: 'If Reagan doesn't win, Anderson will win'.

(viii)(Better): We don't know yet who won. A: 'If a Republican won, then if Reagan didn't win, Anderson won'; B: 'A Republican won'; C: 'If Reagan didn't win, Anderson won'.

3. Possible reactions to these counterexamples

(Note that one could react differently to different examples.)

(a) We're just making a mistake when we judge that A is true in these examples, or when we judge that C is false.

(b) So much the worse for Modus Ponens! Challenge: why does it work when it works?

(c) Give up on the idea that conditionals can be true or false. Challenge: is there some other way of making sense of 'valid' in such a setting? Does Modus Ponens come out 'valid' in the new sense, or not?

(d) "Syntactic mismatch": the trios aren't really of the relevant syntactic forms at all—i.e. despite appearances, B and C do not occur as *constituents* in A (at least on the natural parsing of A).

1. *First version:* wide-scoping. E.g.: the logical form of 'If it's Tuesday we must be in Paris' is 'Must [[if it's Tuesday] [we be in Paris]]', or maybe 'We must [[be in Paris] [if it's Tuesday]]'. (Or something involving 'movement' which gets the same results in a somewhat more principled way.)

   - A very old idea—the medievals had a distinction between *necessitas consequentis* and *necessitas consequentiae* which was basically this. Often combined with an account on which the embedded conditional is material. But this is not compulsory.

2. *Second version:* Kratzerian syntax. E.g. the logical form of 'If it's Tuesday we must be in Paris' is '[Must [if it's Tuesday]] [we be in Paris]'.

(e) "Semantic mismatch": certain key words that appear both in A and C—e.g. 'should', 'likely', the inner 'if' in the McGee example—are ambiguous (or context-sensitive), and are not being interpreted in the same way when we judge A true and C false. If we fix the interpretation of these words in C to be the same one that's natural for A, then C is true.

Syntactic mismatch and semantic mismatch

Conditionals

30 November 2017

1. Recap: putative counterexamples to Modus Ponens

(i) The paradox of gentle murder (Forrester): A:'If John kills his mother, he ought to kill her gently'. B:'John will kill his mother'. C:'John ought to kill his mother gently'.

(ii) (A slight variant:) A 'If we're going to leave tomorrow we should start packing now'; B: 'We're going to leave tomorrow'; C: 'We should start packing now'.

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(v) (A slight variant): A: 'If it's Tuesday we must be in Paris'; B: 'It's Tuesday'; C: 'We must be in Paris'.

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(viii)(Better): We don't know yet who won. A: 'If a Republican won, then if Reagan didn't win, Anderson won'; B: 'A Republican won'; C: 'If Reagan didn't win, Anderson won'.

2. Two possible reactions to discuss today

(a) "Syntactic mismatch": the trios aren't really of the relevant syntactic forms at all—i.e. despite appearances, B and C do not occur as *constituents* in A (at least on the natural parsing of A).

   1. *First version:* wide-scoping. E.g.: the logical form of 'If it's Tuesday we must be in Paris' is 'Must [[if it's Tuesday] [we be in Paris]]', or maybe 'We must [[be in Paris] [if it's Tuesday]]'. (Or something involving 'movement' which gets the same results in a somewhat more principled way.)
   2. *Second version:* Kratzerian syntax. E.g. the logical form of 'If it's Tuesday we must be in Paris' is '[Must [if it's Tuesday]] [we be in Paris]'.

(b) "Semantic mismatch": certain key words that appear both in A and C—e.g. 'should', 'likely', the inner 'if' in the McGee example—are ambiguous (or context-sensitive), and are not being interpreted in the same way when we judge A true and C false. If we fix the interpretation of these words in C to be the same one that's natural for A, then C is true.

3. More on syntactic mismatch

Challenge for wide-scopers: tell a story about the meaning of the embedded 'if' sentence and a story about the meaning of the modal (as a unary operator) that together yield plausible truth conditions for the A sentence.

Challenge for Kratzerians: tell a story about the meaning of the modal (as a binary operator) that yields plausible truth conditions for the A sentence.

Contrasts between the the wide-scoping response and the Kratzerian response.

- The wide-scoping response is particularly implausible in the case of the McGee example. Here, I suppose, it would go along with the thought that the real logical form of 'If a Republican won, then if Reagan didn't win Anderson won' is '[If Reagan didn't win] [[if a Republican won] [Anderson won]]'. But how could it be mandatory to reorder the 'if' clauses in this way? And if it's not mandatory, what could make it the preferred diambiguation in the circumstances?
- Kratzer says that her view allows "stacking" of 'if'-clauses, so that 'Modal if P, then if Q, then R' means the same as 'Modal if P and Q then R'. Presumably her favoured logical form is '[[Modal [if P]] [if Q]] R'. But she doesn't—at least in the paper we looked at—offer a story about how this actually comes about, compositionally.
- The following argument poses a special challenge for those want to solve the minders puzzle by wide-scoping:

  (1) Should [if the miners are in shaft 1, we flood only one shaft]
  (2) Should [if the miners are not in shaft 1, we flood only one shaft]
  (3) Should [[if the miners are in shaft 1, we flood only one shaft] and [if the miners are not in shaft 1, we flood only one shaft]]
  (4) Should [we flood only one shaft]

- While the miners puzzle is not so challenging for Kratzerian, it it does put pressure on Kratzer's own, Lewis-inspired, theory about the semantic values for modals (functions from propositions to functions from propositions to propositions).
  - On Kratzer's general theory, modals have two parameters of context-sensitivity: a *modal base* and and *ordering source*. We don't need to worry about what these are: what matters is that the modal base determines an *accessibility relation* R among possible worlds, and the ordering source determines a three-place relation $w_1 \leq_{w_2} w_3$ among possible worlds.

- (Glossing over a certain complexity): given R and ≤, when M is a *necessity modal*, M (P) (Q) is true at w iff either no P-world bears R to w, or some P-world v bears R to w and is such that for every P-world u which bears R to w and is such that $u \leq_w v$, u is a Q-world.
- Kratzer suggests that deontic modals like 'should' are necessity modals where the modal base is what's "practically possible" and the ordering source is a matter of what's best (by relevant standards). So ⟦Should⟧ (P) (Q) is (roughly) true iff either P is practically impossible, or there's a practically possible P world such that all practically possible P-worlds that are *at least as good as it* are Q-worlds.]
- This validates the inference from (1) and (2) to (4) above, despite the misleading structure! No matter what the modal base and ordering source are for a necessity modal □, if □ (P) (Q) and □ (¬P) (Q) are both true, □ (⊤) (Q) is also true.

4. Similar examples involving 'or' rather than 'if'

(5) Either Bob is at home or he must be working
(6) Either Bob is at home or he might be working
(7) Either this is a zebra or it's probably a donkey
(8) Either we're not going to leave tomorrow or we should start packing now

- Kratzerianism doesn't help with these at all.
- Wide-scoping is much more syntactically implausible for these sentences. Also, wide-scoping seems to give wrong truth conditions for (6) and (7).
- Insofar as they seem to involve the same phenomenon as the earlier conditional cases, they tend to undermine the use of it to support any special view about conditionals (unless it goes along with some relevantly similar view about 'or').

5. More on semantic mismatch

The proponent of semantic mismatch thinks that in the examples, the most natural readings for the embedded modals (and conditionals? and attitude verbs?) embedded in the consequents of conditionals is different from the most natural readings for the same words when unembedded.

Q1: is there a *possible* reading of the unembedded modal sentence that's the same as the reading that's natural for the consequent of the conditional?

Q2: is there a *possible* reading of the conditional where its consequent has the reading that's natural for the unembedded modal sentence?

- If the answer to both questions was 'no', it's unclear what we even mean by 'valid' for the relevant arguments. Any argument that doesn't *have* any uniform interpretations will, vacuously, preserve truth on all uniform interpretations!

- Some evidence that the answer to Q2 is yes: '[Even] If John is robbing a bank, he shouldn't be robbing a bank'. '[Even] if John is [in fact] robbing a bank, he should be spending his life in Africa helping the poor'.
- Some evidence the answer to Q1 is yes: 'Either it's a donkey or it's likely to be a mule. There's a good chance that it's not a donkey. So, there's a good chance that it's likely to be a mule.'

A view where the answer to both questions is 'yes':

(i) Model *questions* as *partititions of logical space:* sets of propositions (the *answers* to the question) such that exactly one of them is true at each world. Yes-no questions are two-cell partitions.

(ii) At any point in a conversation certain questions are "under discussion". Utterance of a sentence generally puts under discussion both the yes-no question corresponding to the entire sentence, and the questions corresponding to its parts.

(iii)In general, every modal and conditional is associated with an accessibility relation.

(iv)Often (typically?), the accessibility relation for a modal or conditional uttered when certain questions Q_1, Q_2,… are under dicussion is *constrained by* Q_1, Q_2, … — i.e. it's such that for v to be accessible from w, v and w must belong to the same cell of Q_1, and belong to the same cell of Q_2, and…

- E.g.: 'I'm pretty confident that this coin might land Tails', said of a coin chosen from a bucket with many ordinary coins and some double-headed coins. QUD: 'Is the coin double headed?'
- When 'might' is constrained by the yes-no question 'A?', 'might P' is equivalent to '(A and might (A and P)) or (not-A and might (not-A and not P)).
- When 'if' is constrained by 'A?', 'if P, Q' is equivalent to '(A and if A and P, Q) or (not-A and if not-A and P, Q)'.

(v) When we are interpreting B in 'If A, B', or 'A or B' or 'A and B', the yes-no question corresponding to A will typically be under dicussion, and any modals that occur in B will be constrained by this question.

- When this happens, 'If A, might P' will be equivalent to 'If A, (A and might (A and P)) or (not-A and might (not-A and not P))', which is in turn logically equivalent to 'If A, might (A and P)'.
- 'If A, (if P, Q)' will be equivalent to 'If A, (A and if A and P, Q) or (not-A and if not-A and P, Q)', which is in turn logically equivalent to 'If A, (if A and P, Q)'.

Alternative explanations of the phenomena that motivate Kratzer

Conditionals

30 November 2017

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- Some evidence that the answer to Q2 is yes: '[Even] If John is robbing a bank, he shouldn't be robbing a bank'. '[Even] if John is [in fact] robbing a bank, he should be spending his life in Africa helping the poor'.
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(i) Model *questions* as *partititions of logical space:* sets of propositions (the *answers* to the question) such that exactly one of them is true at each world. Yes-no questions are two-cell partitions.

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- When 'might' is constrained by the yes-no question 'A?', 'might P' is equivalent to '(A and might (A and P)) or (not-A and might (not-A and not P)).
- When 'if' is constrained by 'A?', 'if P, Q' is equivalent to '(A and if A and P, Q) or (not-A and if not-A and P, Q)'.

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- When this happens, 'If A, might P' will be equivalent to 'If A, (A and might (A and P)) or (not-A and might (not-A and not P))', which is in turn logically equivalent to 'If A, might (A and P)'.
- 'If A, (if P, Q)' will be equivalent to 'If A, (A and if A and P, Q) or (not-A and if not-A and P, Q)', which is in turn logically equivalent to 'If A, (if A and P, Q)'.

2. Adverbial quantifiers versus nominal quantifiers

Adverbial:

(1) a. Always[/Mostly/Sometimes] if a student goofs off, he will fail
b . Always[/Mostly/Sometimes] a student will fail if he goofs off
c. Always[/Mostly/Sometimes] if a student goofed off, he failed
d. Always[/Mostly/Sometimes] a student failed if he goofed off

Nominal

(2) a. Every student[/no student] will fail if he goofs off
b. Most students[/some students] will fail if they goof off
c. Every student[/no student] failed if he goofed off
d. Most students[/some students] failed if they goofed off

What's the relation between these and

(3) a. Every student/no student who goofs off will fail
b. Most students/some students who goof off will fail
c. Every student/no student who goofed off failed
d. Most students/some students who goofed off failed

Kai von Fintel and Sabine Iatridou note a semantic difference between (2ab) and (3ab): (2b) could be true even if in fact it makes almost everyone work hard except for the teacher's pet who won't fail no matter what.

- Is there the same difference between (1ab) and (3ab)?

- Is there a similar difference between (2cd) and (3cd)?

Cases where an 'if' clause seems interchangeable with a normal relative clause:

(4) a. Every query was answered if it came from a reputable address.
b. Every day, he went for a walk after lunch if he had time.
c. Every participant chose A if given a choice between A and B.

(5) a. Every query which came from a reputable address was answered.
b. Every day on which he had time, he went for a walk after lunch.
c. Every participant who was given a choice between A and B chose A.

3. Syntactic high-jinks for 'if' under nominal quantifiers?

Could the logical form of (3c) be like this?

?? [Every [[student] [if he goofed off]]] [failed]

4. Read the conditional as material?

(6) a. No query went unanswered if it came from a reputable address.
b. Most queries were answered if they came from a reputable address.

(7) a. No query which came from a reputable address went unanswered.
b. Most queries which came from a reputable address were answered.

5. Appeal to contextual domain restriction?

(8) Every query was answered if it came from a reputable address and discarded if it came from a particularly disreputable address

(9) Every query which came from a reputable address was answered and every query which came from a particularly disreputable address was discarded

6. Can a domain restriction solution also work for adverbial quantifiers?

Adverbial quantiers are often restricted in a way that's influenced (pragmatically?) by material in the sentence:

(10) Tai always eats with chopsticks
(11) Officers always had lunch with ballerinas
(12) Leopards usually attack monkeys in trees